

# Physical Chemistry (I)

## Lecture 24

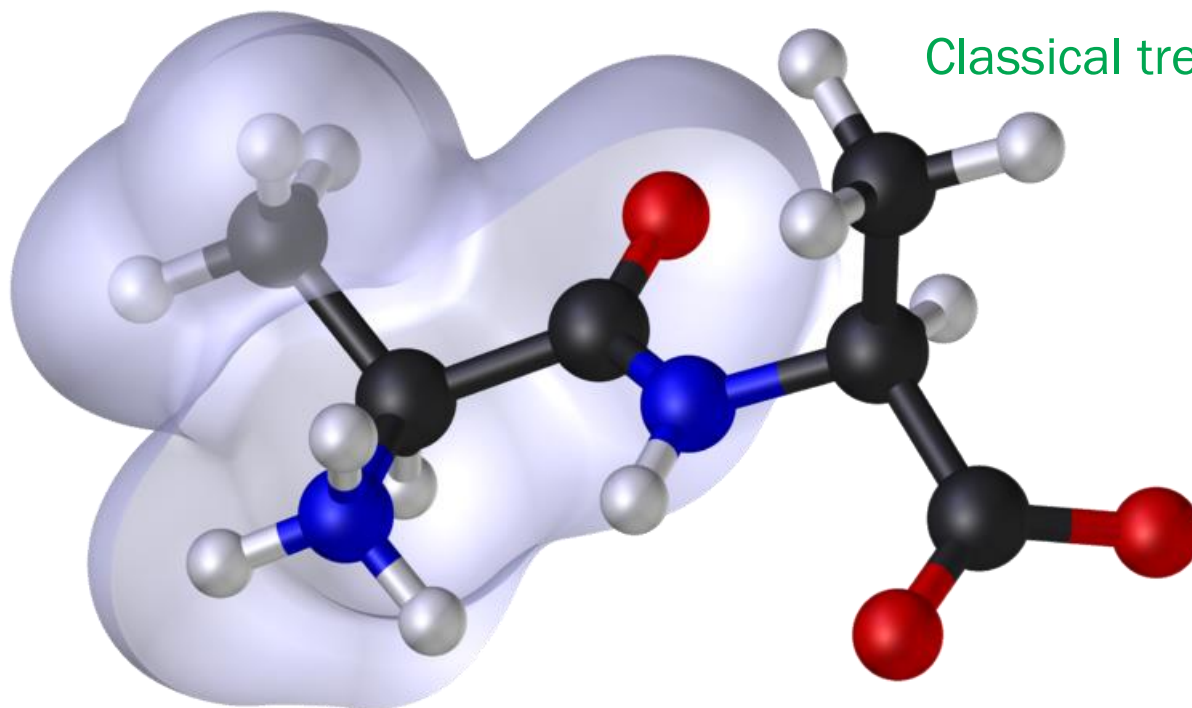
### *Special Lecture on Molecular Modeling*



# Molecular Modeling

Quantum treatment

Classical treatment



# Computational Chemistry

- Quantum treatment

- *ab initio* methods
- semiempirical methods
- Density functional theory (DFT)



Nobel prize in Chemistry 1998

- Classical treatment

- Molecular dynamics (MD)
- Monte Carlo (MC) methods

- Combination: QM/MM, ...

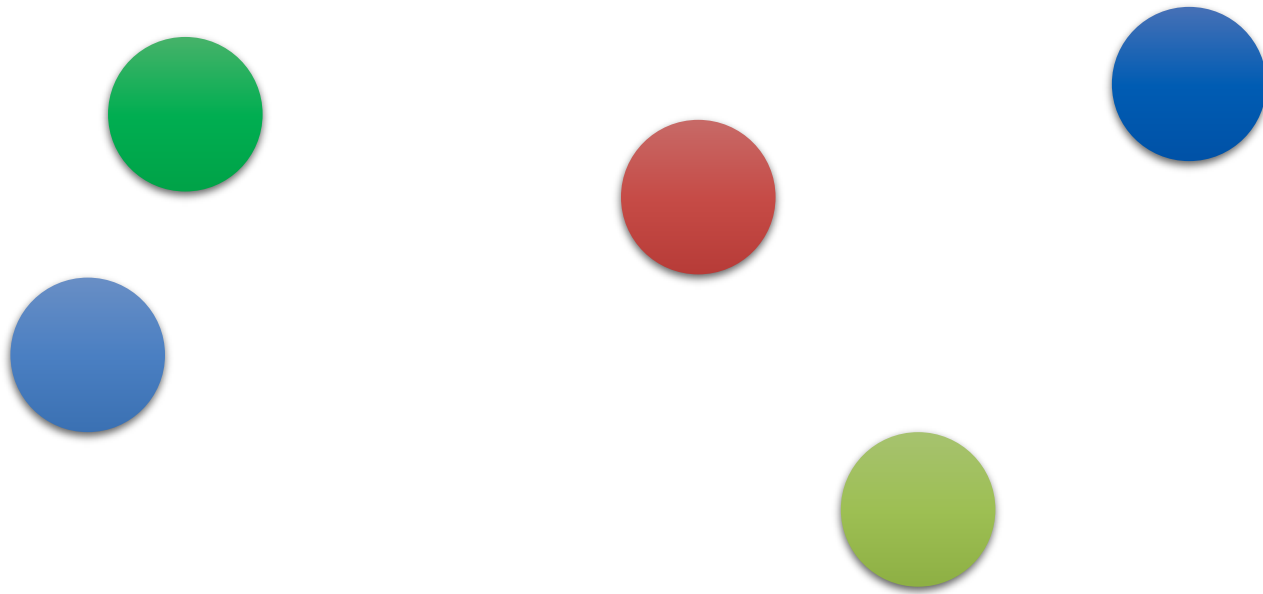


Nobel prize in Chemistry 2013

# 1. Molecular Dynamics

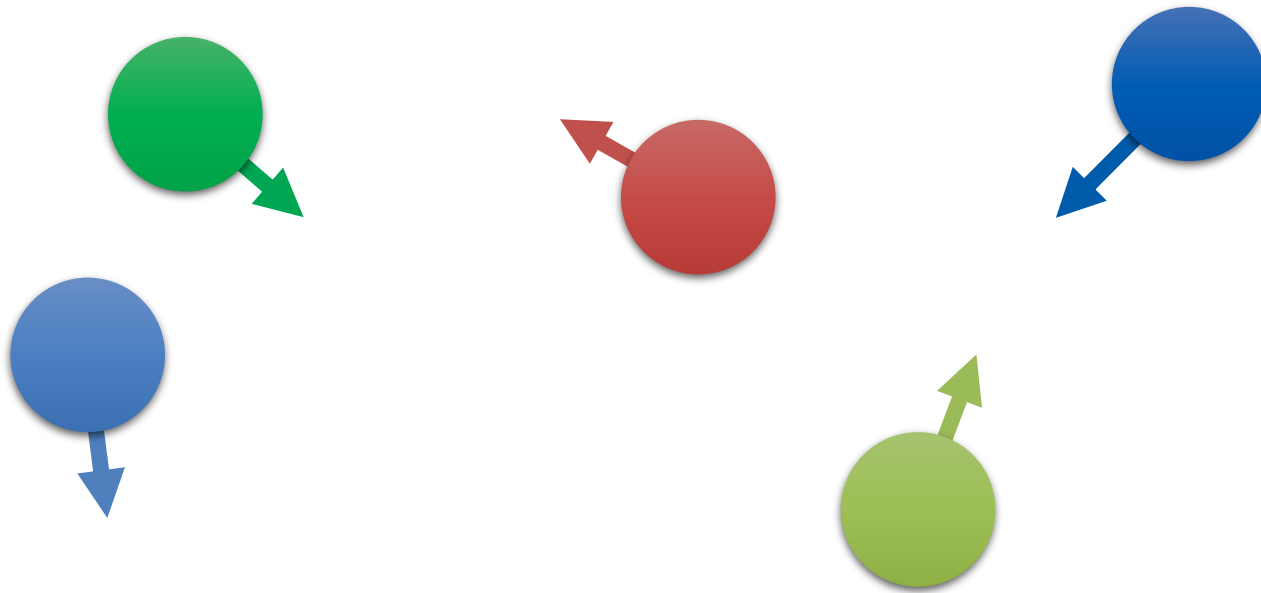


# What is Molecular Dynamics?



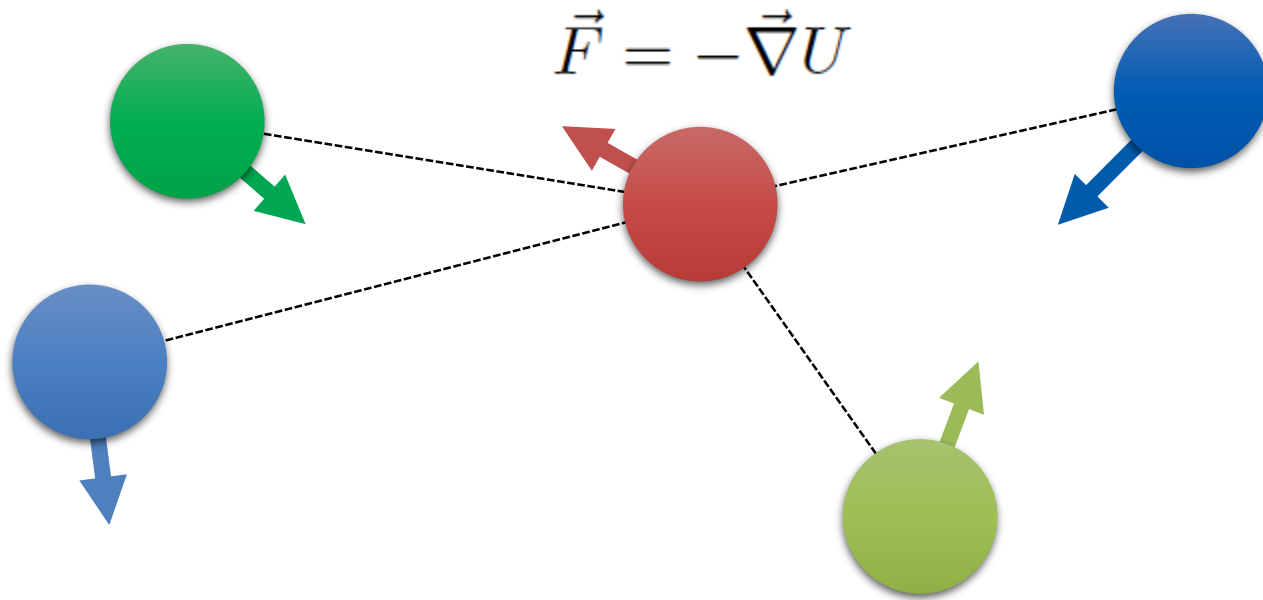
# Molecular Dynamics (MD)

Each particle has **position** and **velocity**



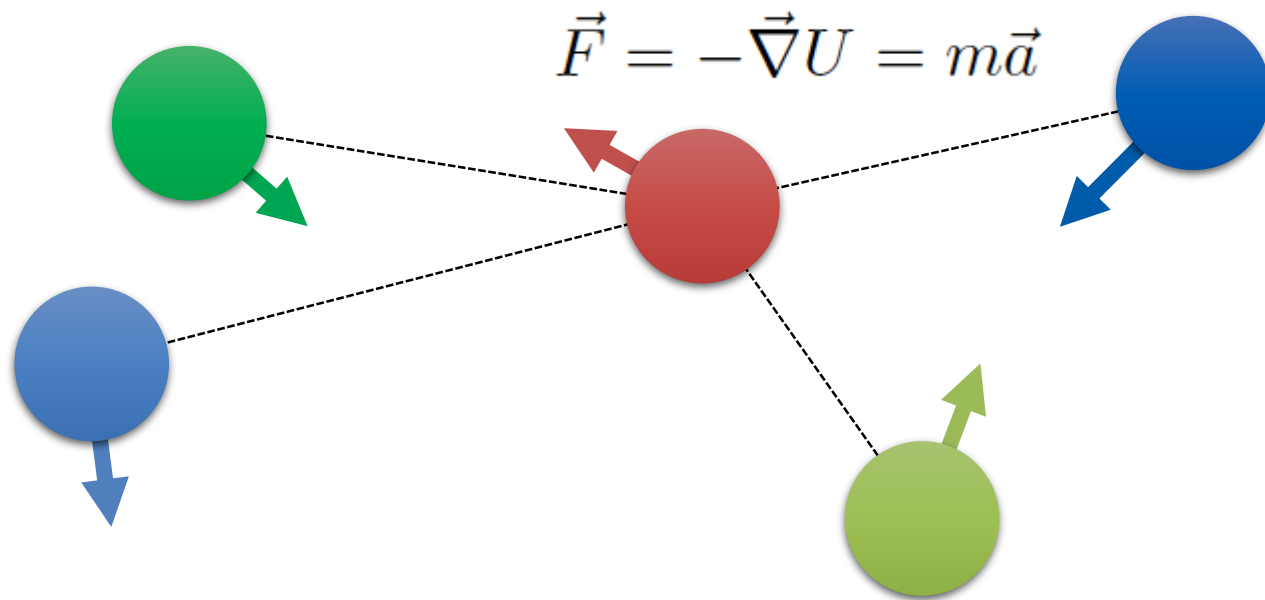
# Molecular Dynamics (MD)

In each MD step,  
calculate **force** on each particle



# Molecular Dynamics (MD)

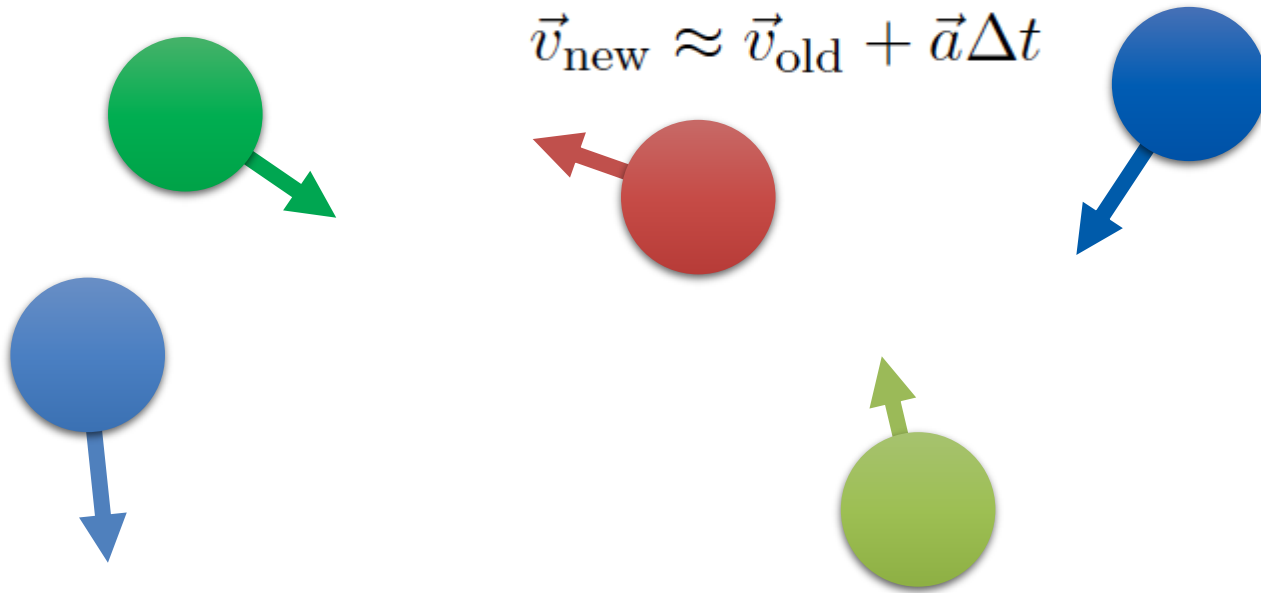
... and calculate **acceleration**



# Molecular Dynamics (MD)

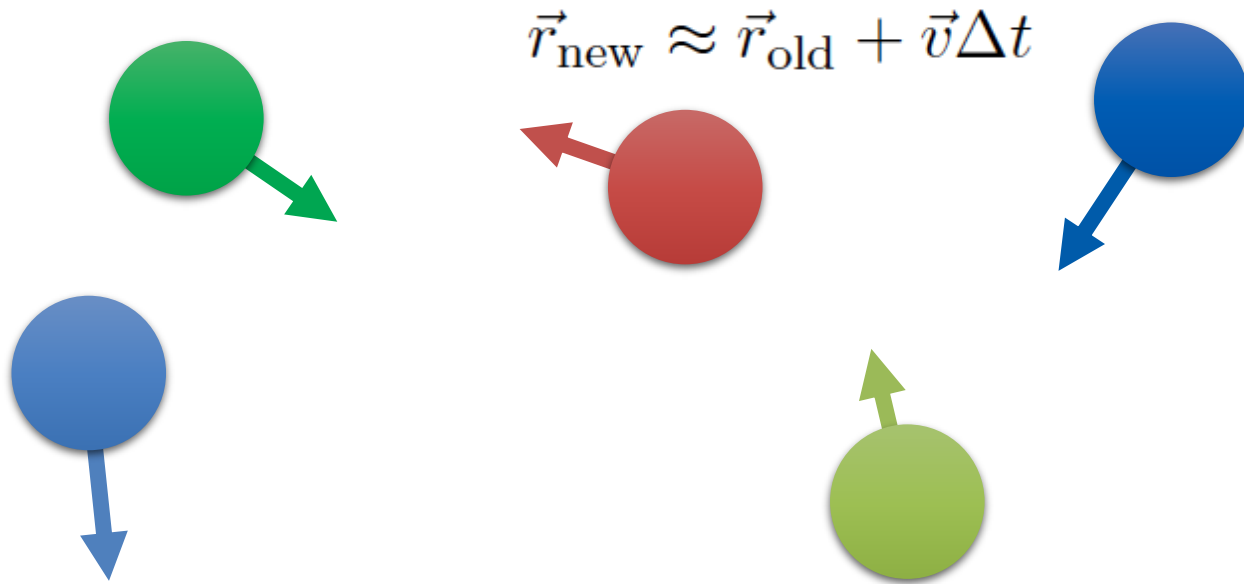
Now change the **velocities**

$$\vec{v}_{\text{new}} \approx \vec{v}_{\text{old}} + \vec{a}\Delta t$$



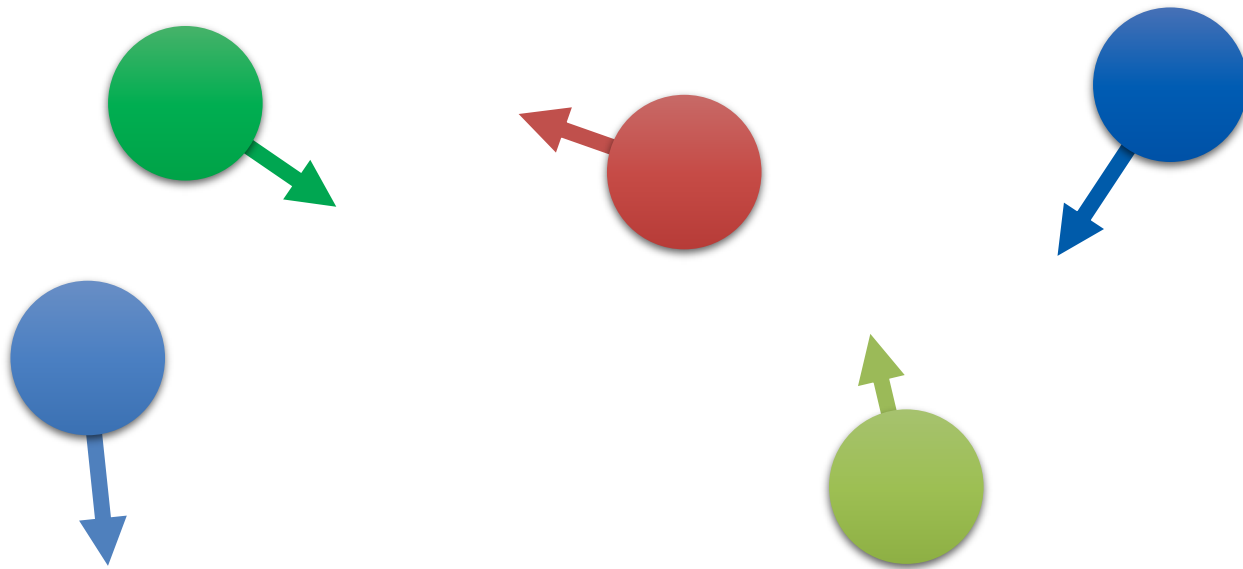
# Molecular Dynamics (MD)

Move the particles accordingly



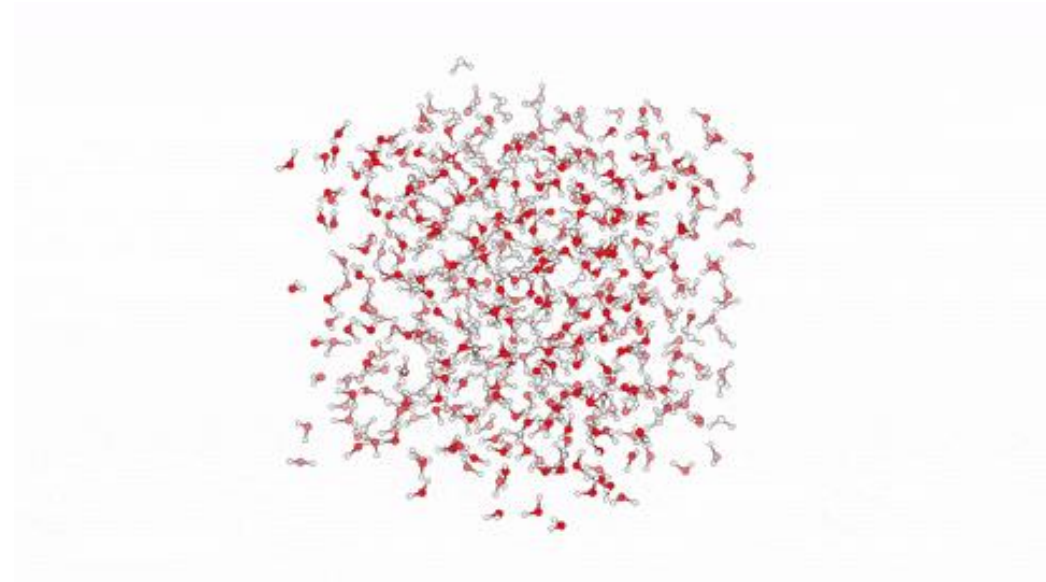
# Molecular Dynamics (MD)

Repeat this again and again



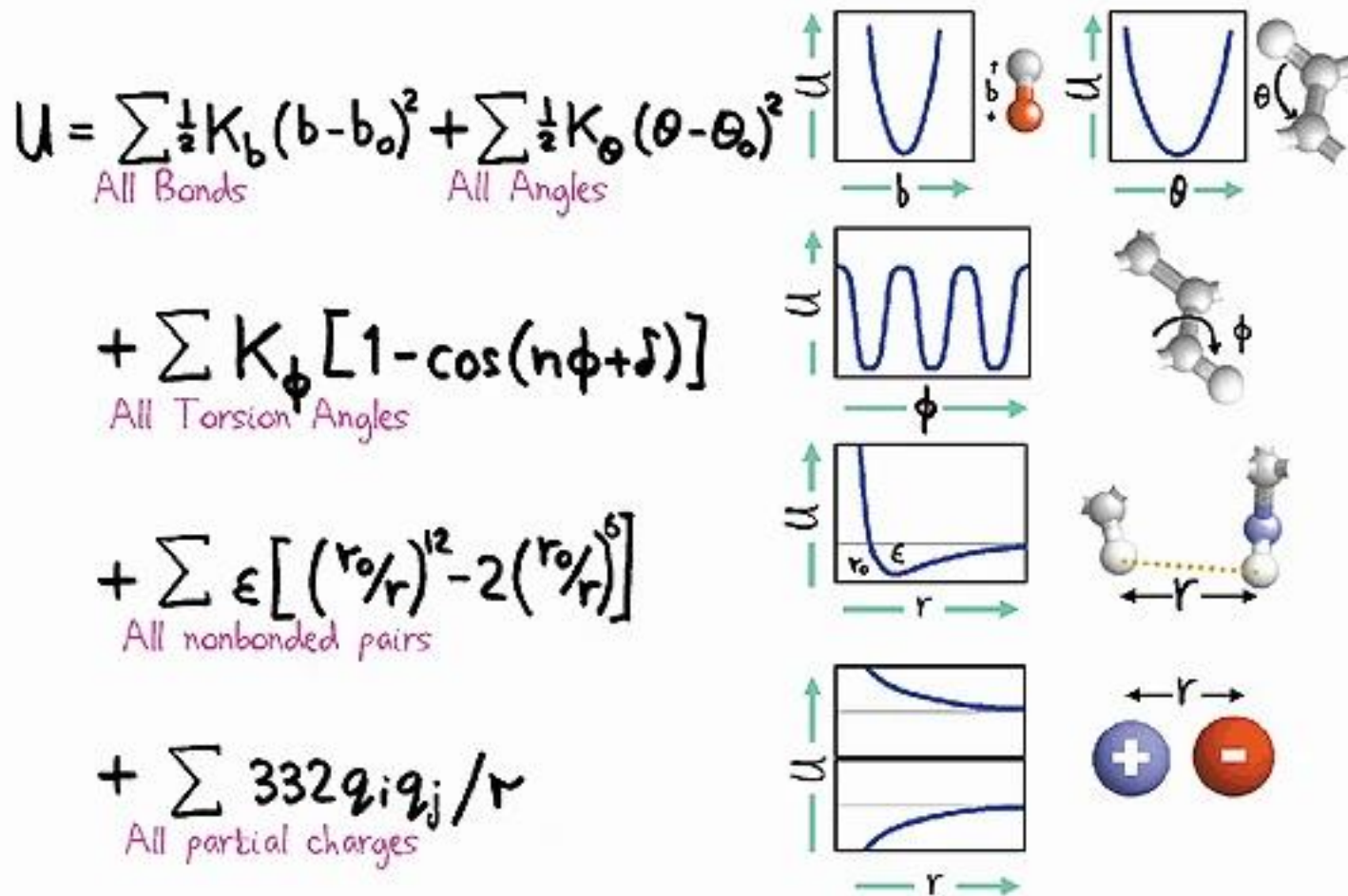
# Ingredients of MD

- Timestep
- Potential energy or interactions
- Initial conditions
  - Positions
  - Velocities



(Animation: [https://upload.wikimedia.org/wikipedia/commons/f/f4/MD\\_water.gif](https://upload.wikimedia.org/wikipedia/commons/f/f4/MD_water.gif))

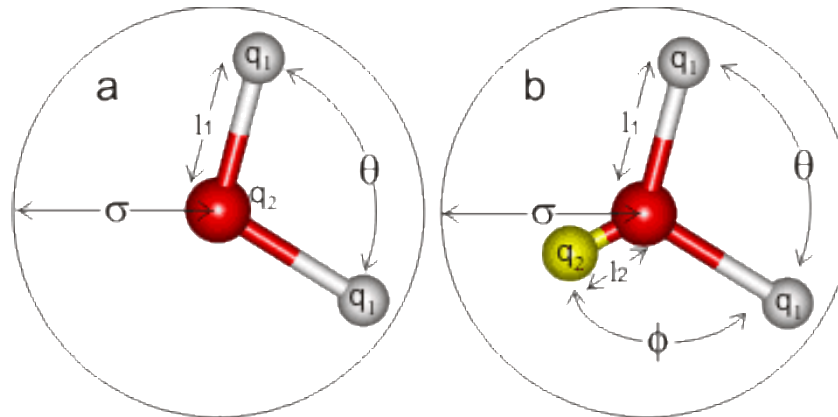
# Potential Energy: “Force Field”



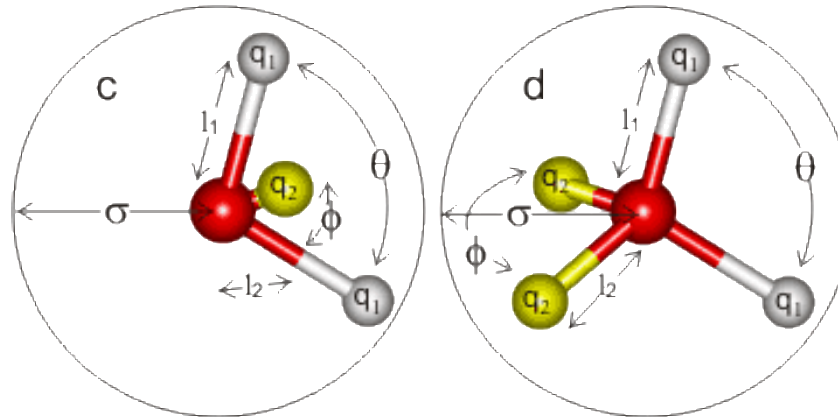
Levitt, *Nat Struct Biol* (2001)

# Water Models

- SPC (1981)
- TIP3P (1983)
- SPC/E (1987)



- TIP4P (1983)



- TIP5P (2000)

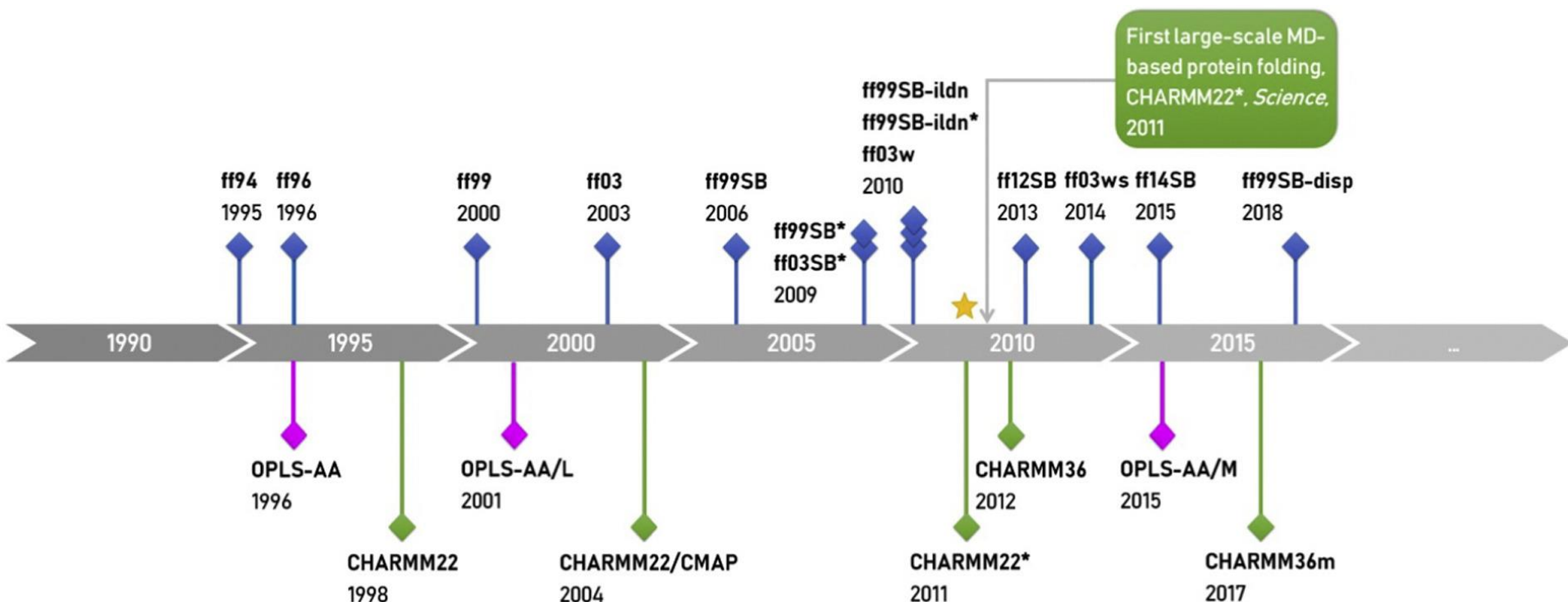
Image: [http://www1.lsbu.ac.uk/water/water\\_models.html](http://www1.lsbu.ac.uk/water/water_models.html)

# Validation of Water Models

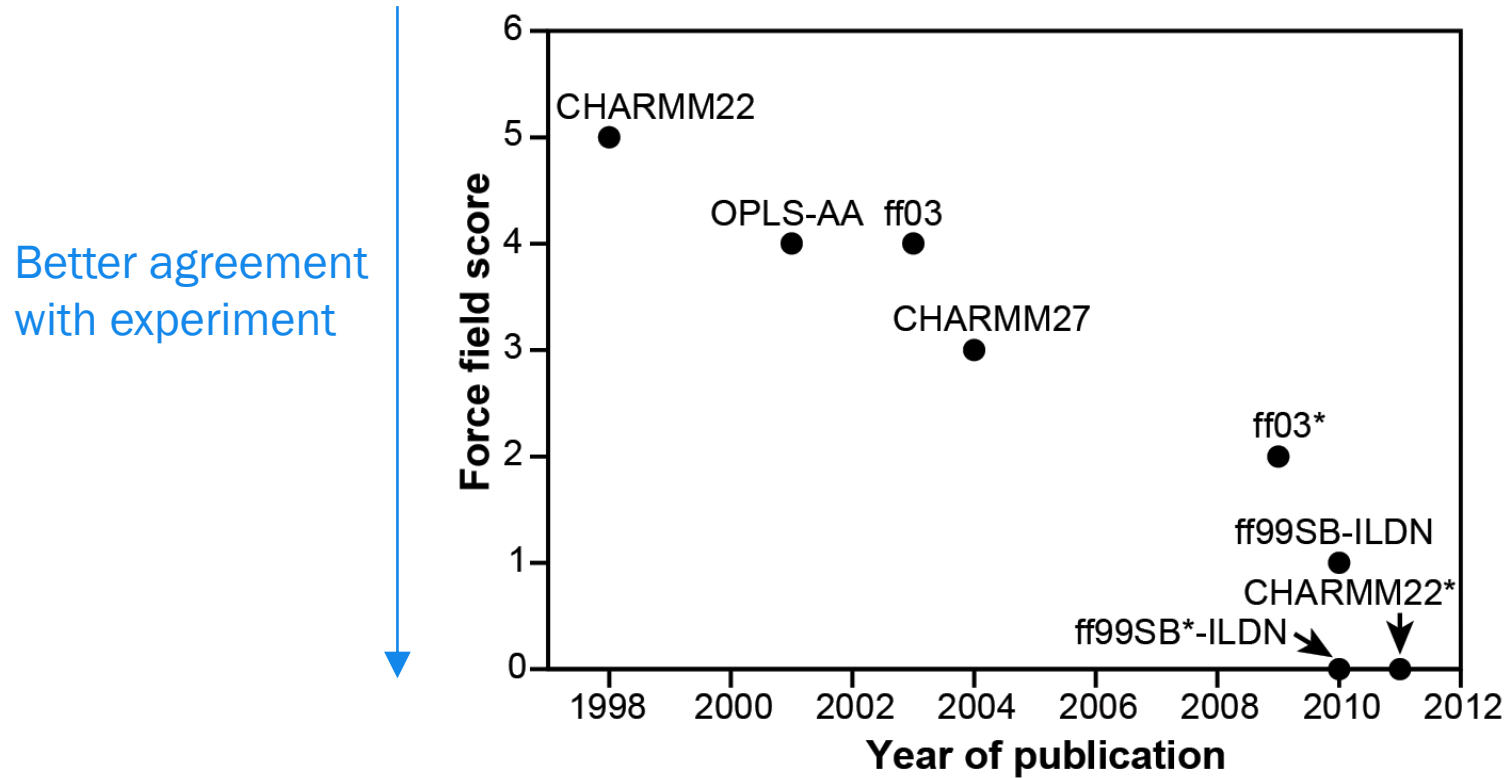
| Model      | % of the experimental value |                 |                      |
|------------|-----------------------------|-----------------|----------------------|
|            | Specific heat, $C_p$        | Shear viscosity | Thermal conductivity |
| SPC        | 102                         | 31              | 144                  |
| SPC/E      | 108                         | 37              | 153                  |
| TIP3P      | 107                         | 36              | 146                  |
| TIP4P      | 118                         | 47              | 135                  |
| TIP4P-Ew   | 115                         | 64              | 147                  |
| TIP4P/2005 | 120                         | 65              | 149                  |
| TIP5P      | 120                         | 88              | 111                  |
| TIP5P-Ew   | 141                         | 91              | 102                  |



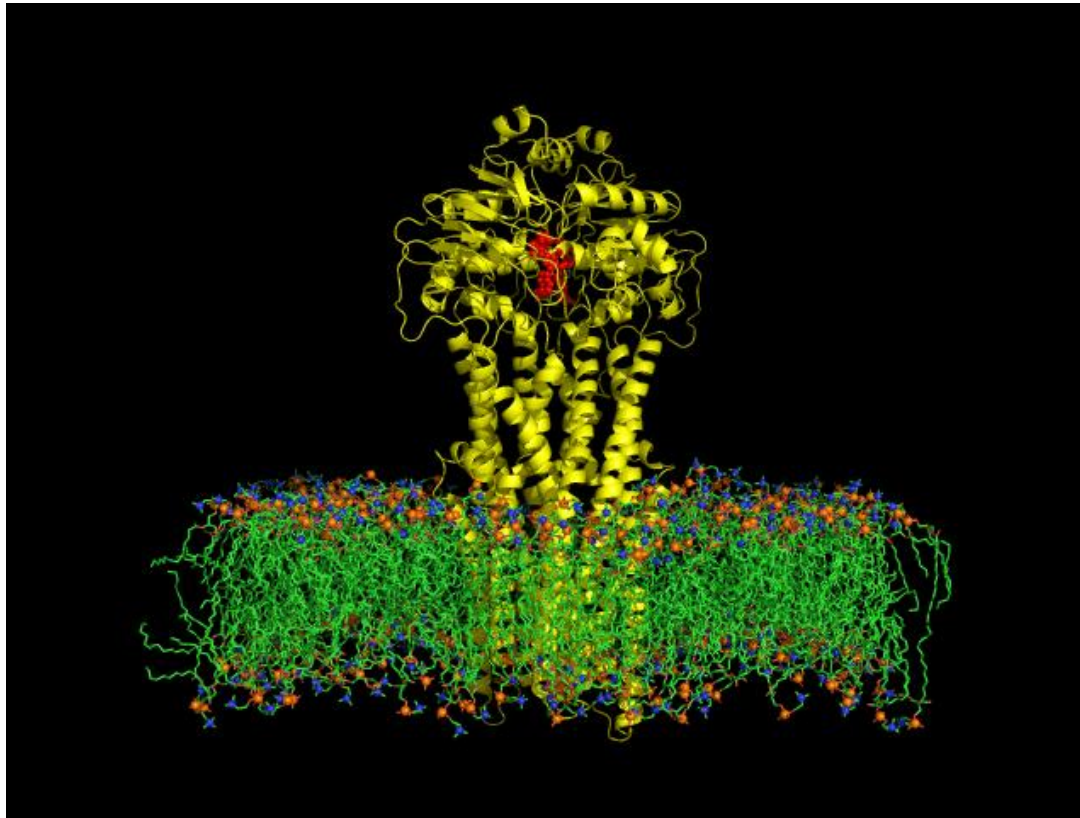
# Popular Force Fields for Proteins



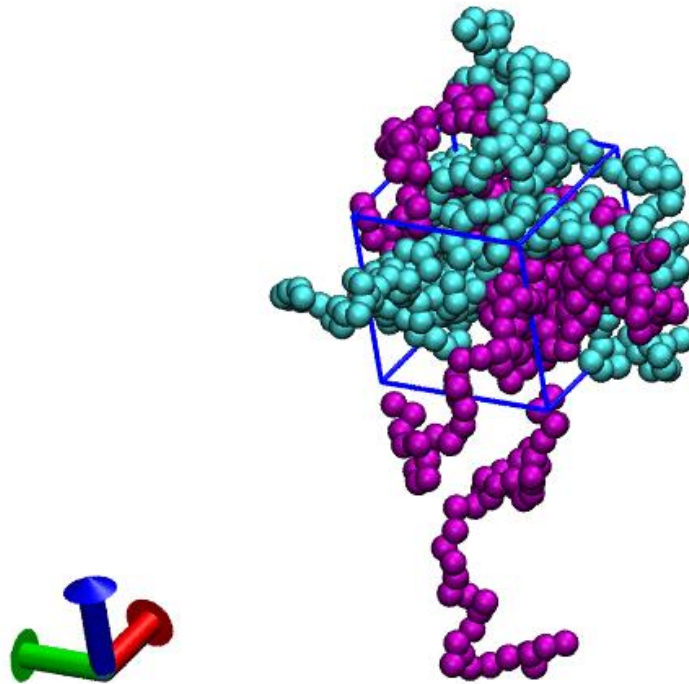
# Validation of Protein Force Fields



# Example: Membrane-Bound Protein



# Example: Polymers



# Example: Solids

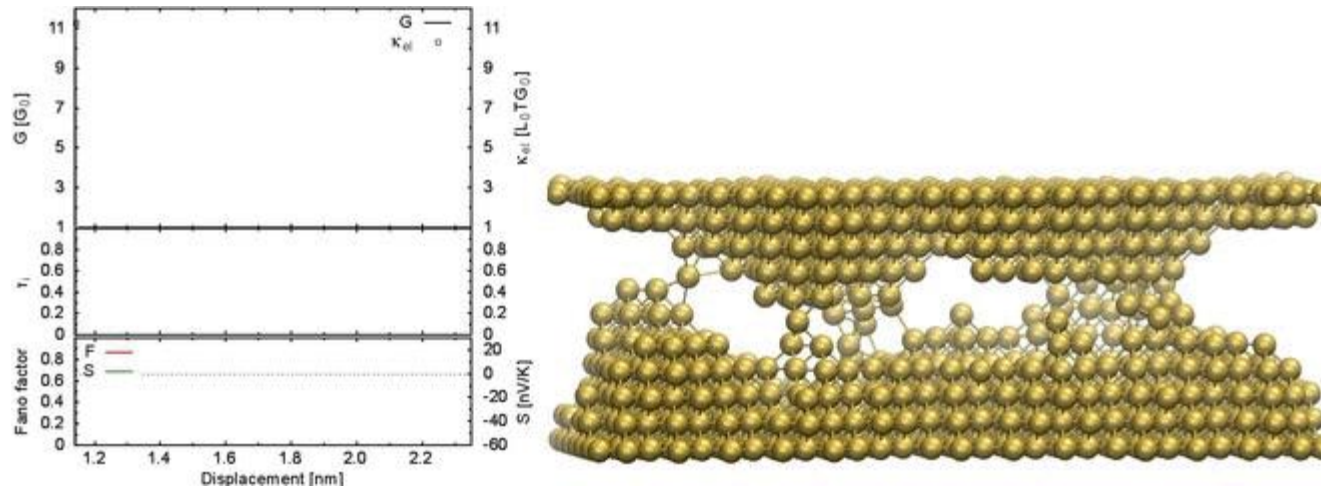


Image: <https://www.nielaba.uni-konstanz.de/en/research/research-projects/nanostructures/atomic-contacts/>

# Energy Conservation

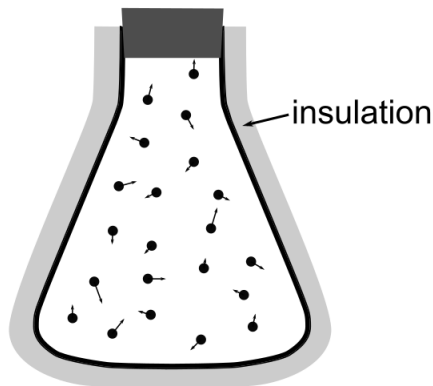
$$\vec{F} = m\vec{a}$$

$$-\nabla E_{\text{pot}} = m \frac{d^2 \vec{r}}{dt^2}$$

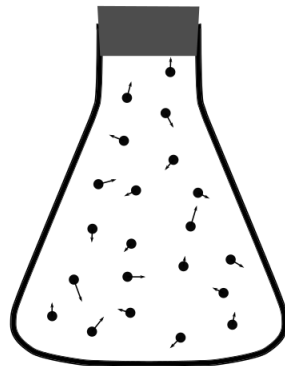
Total system energy is conserved

(isolated system)

# Statistical Ensembles



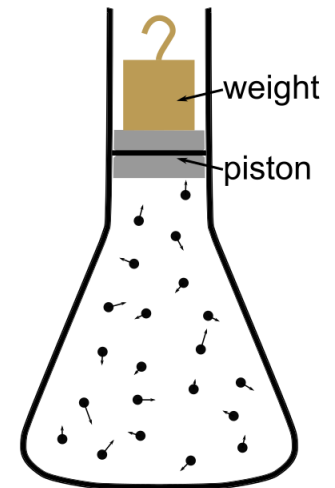
**Microcanonical**  
(const. NVE)



**Canonical**  
(const. NVT)



**Grand Canonical**  
(const.  $\mu$ VT)



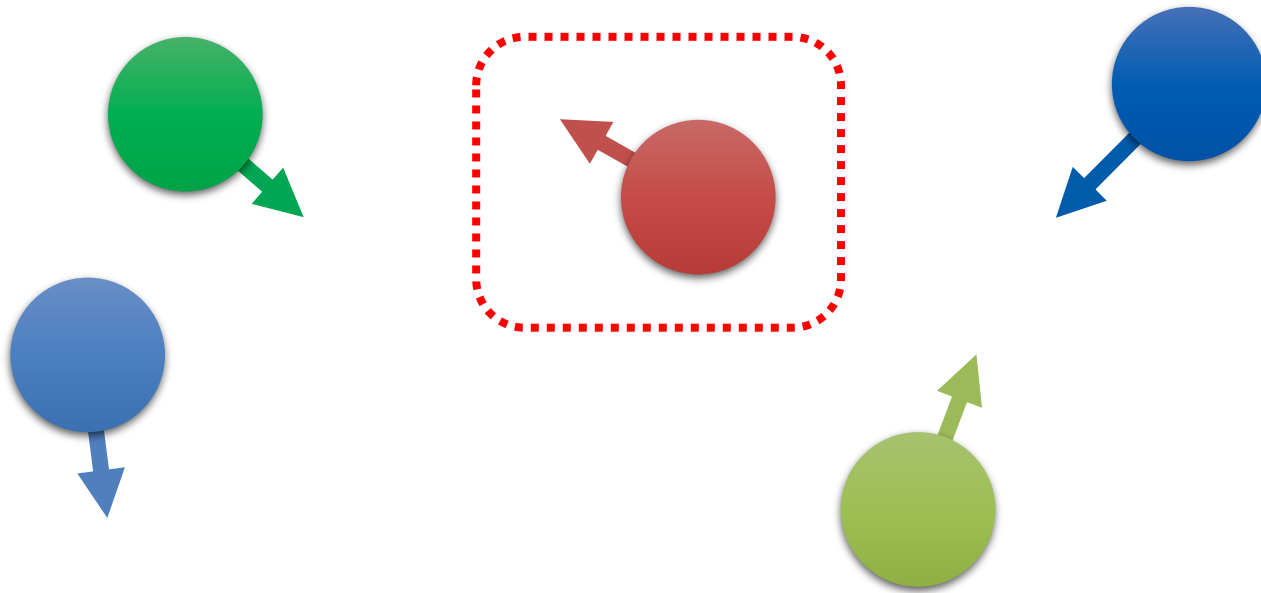
**Gibbs or  
Isobaric-isothermal**  
(const. NPT)

# Thermostats and Barostats

- Thermostats: temperature conserved
  - Andersen thermostat
  - Nosé-Hoover thermostat
  - ...
- Barostats: pressure conserved
  - Berendsen barostat
  - ...

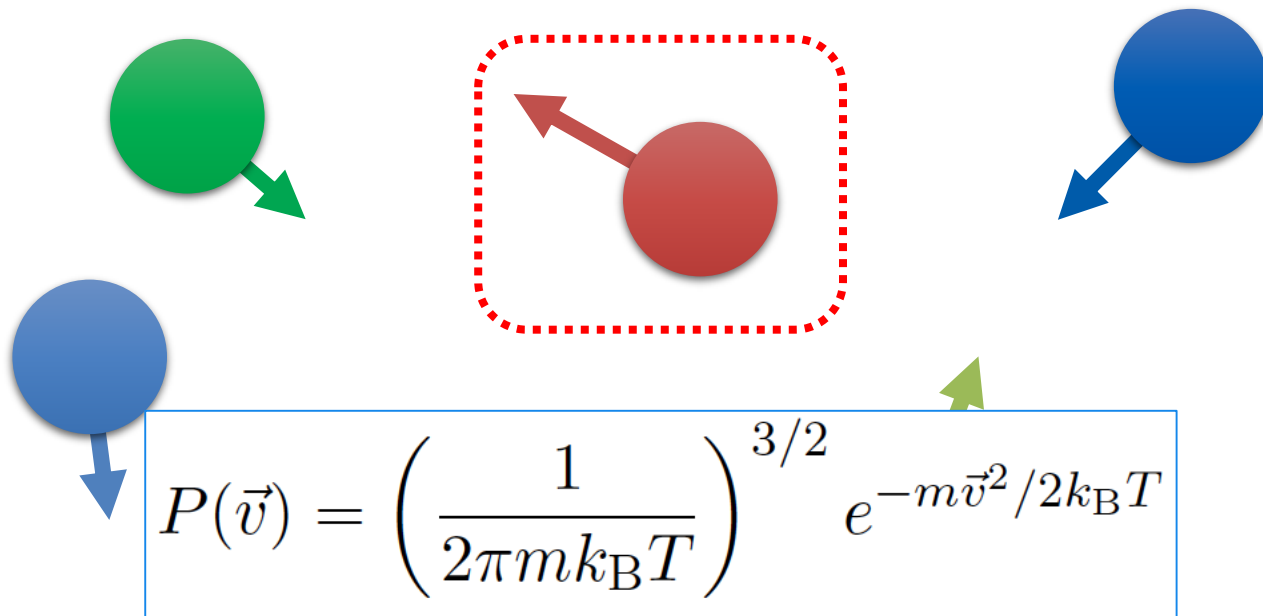
# Andersen Thermostat

Randomly choose a particle (Poisson process)



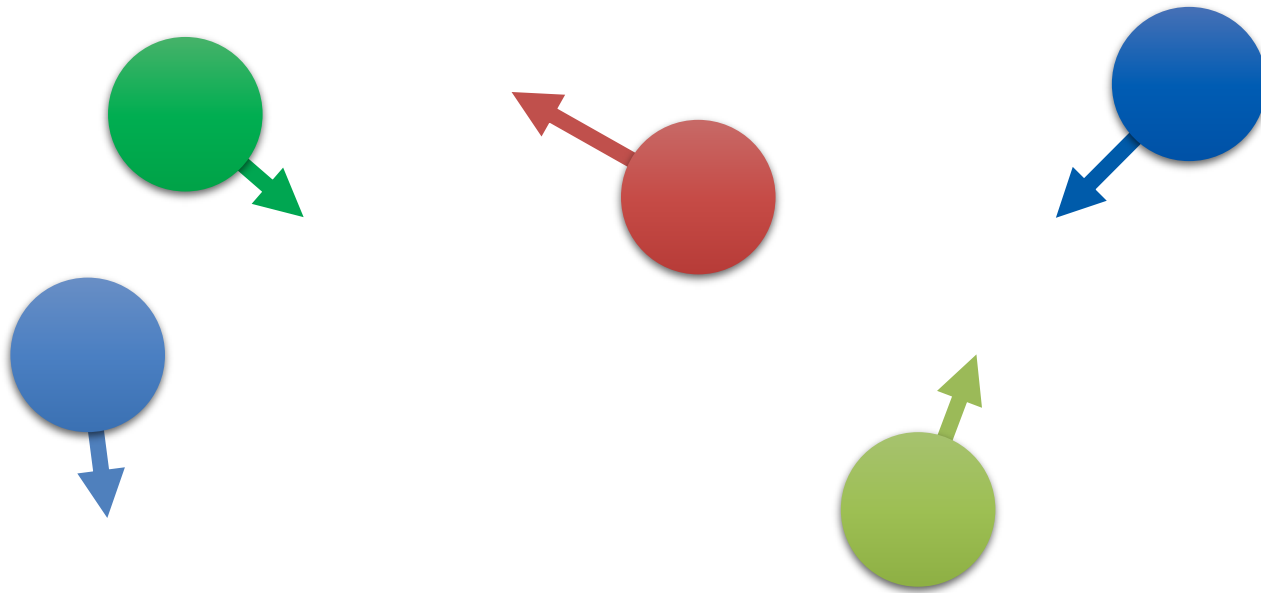
# Andersen Thermostat

Reset its velocity to the one drawn from the [Maxwell-Boltzmann distribution](#)

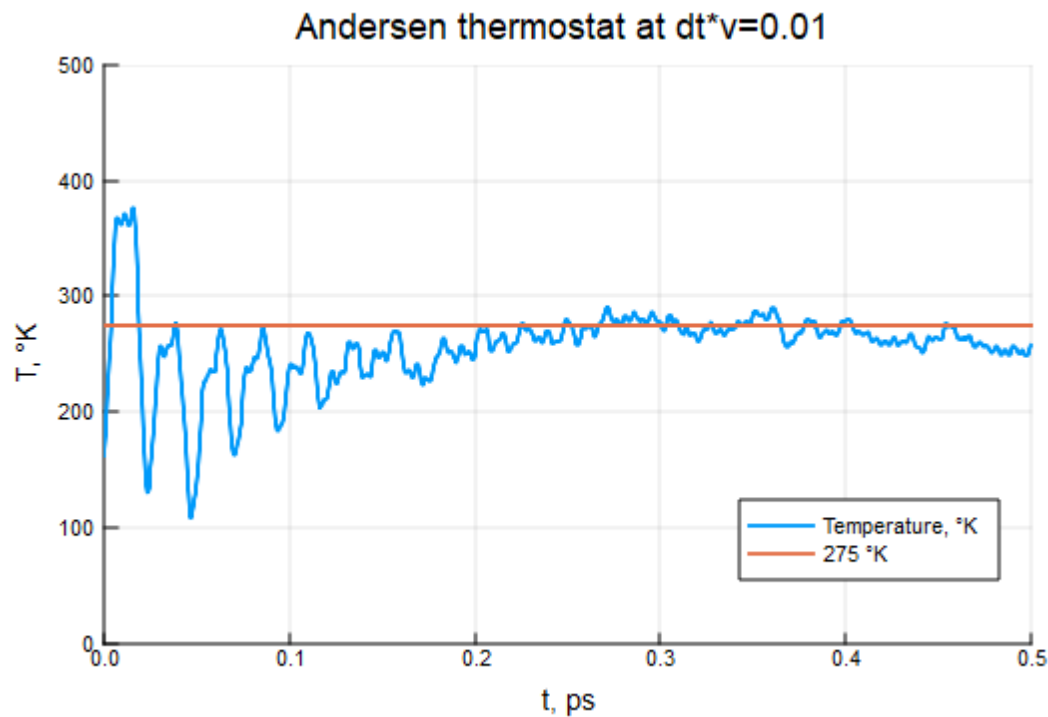


# Andersen Thermostat

Repeat this again and again

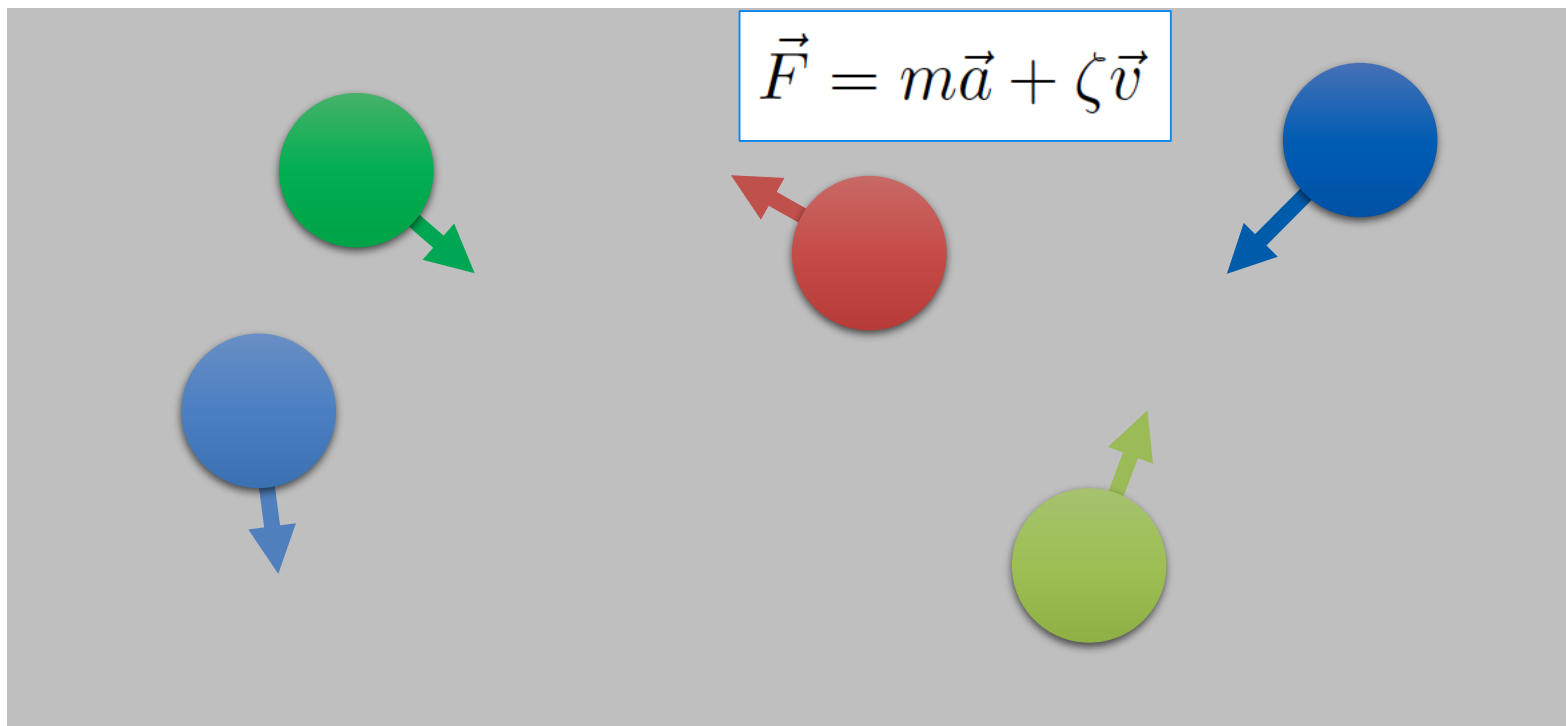


# Andersen Thermostat



# Nosé-Hoover Thermostat

Add **frictional** medium



# Nosé-Hoover Thermostat

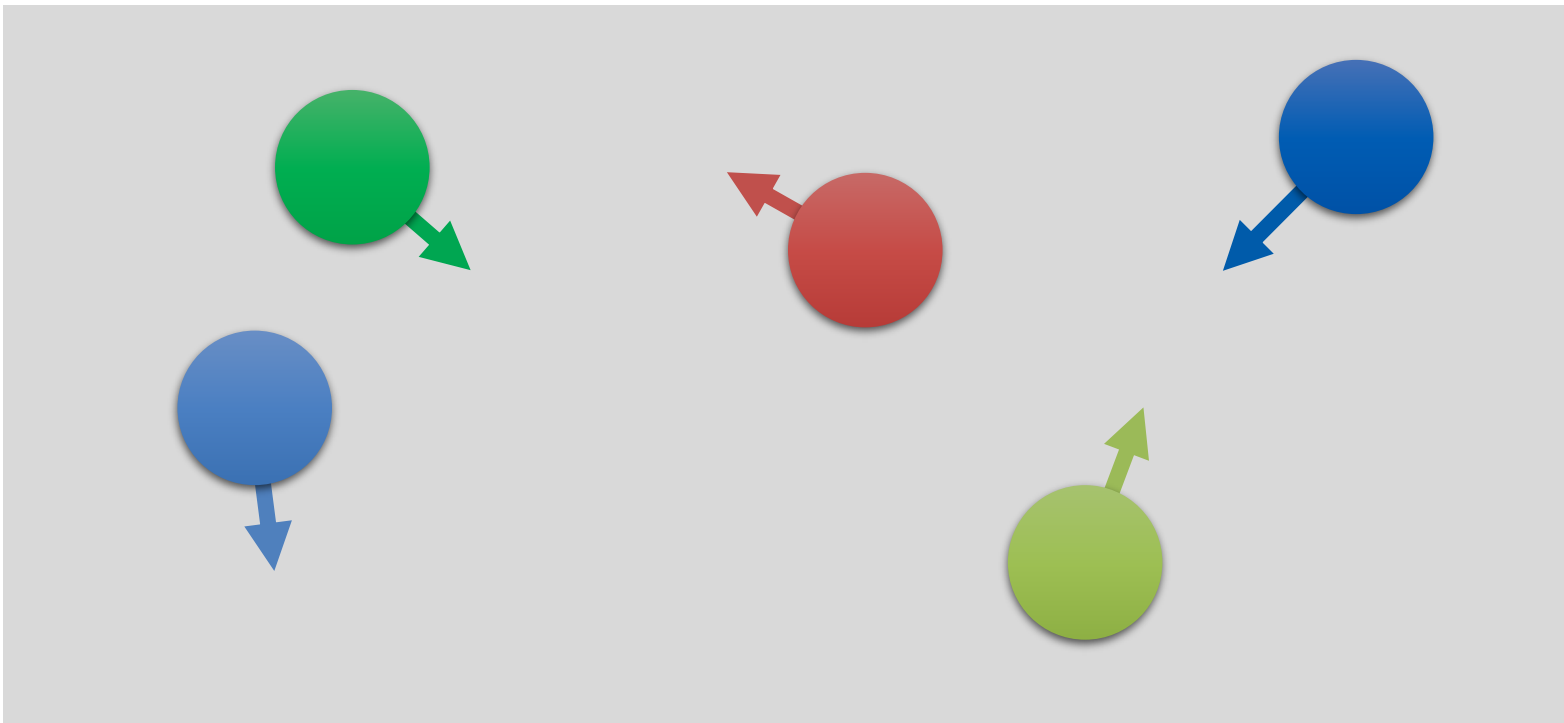
Frictional force depends on the difference between total energy and target energy

$$\vec{F} = m\vec{a} + \zeta\vec{v}$$

$$\frac{d\zeta}{dt} \propto \sum_i \frac{1}{2} m_i \vec{v}_i^2 - \frac{3}{2} N k_B T$$

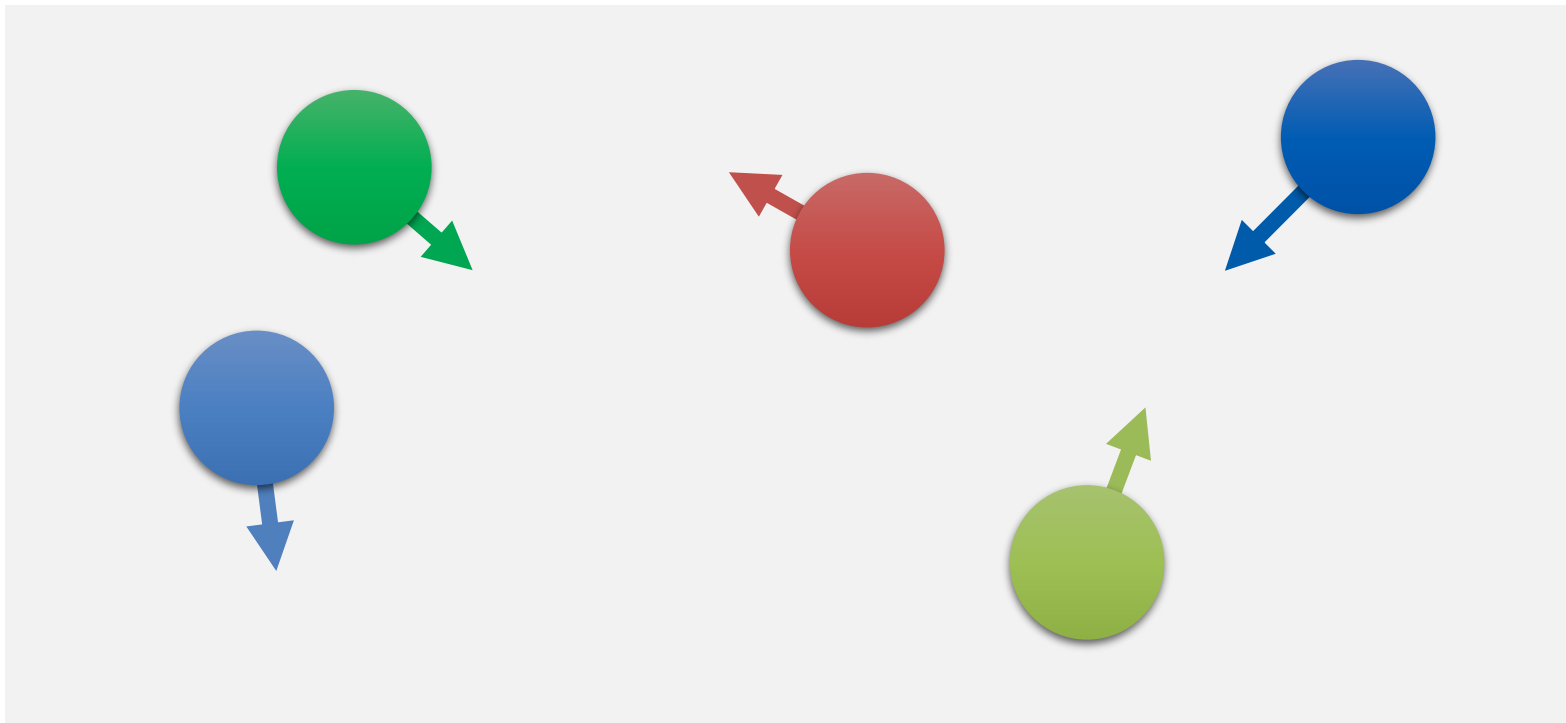
# Nosé-Hoover Thermostat

As the simulation goes on ...



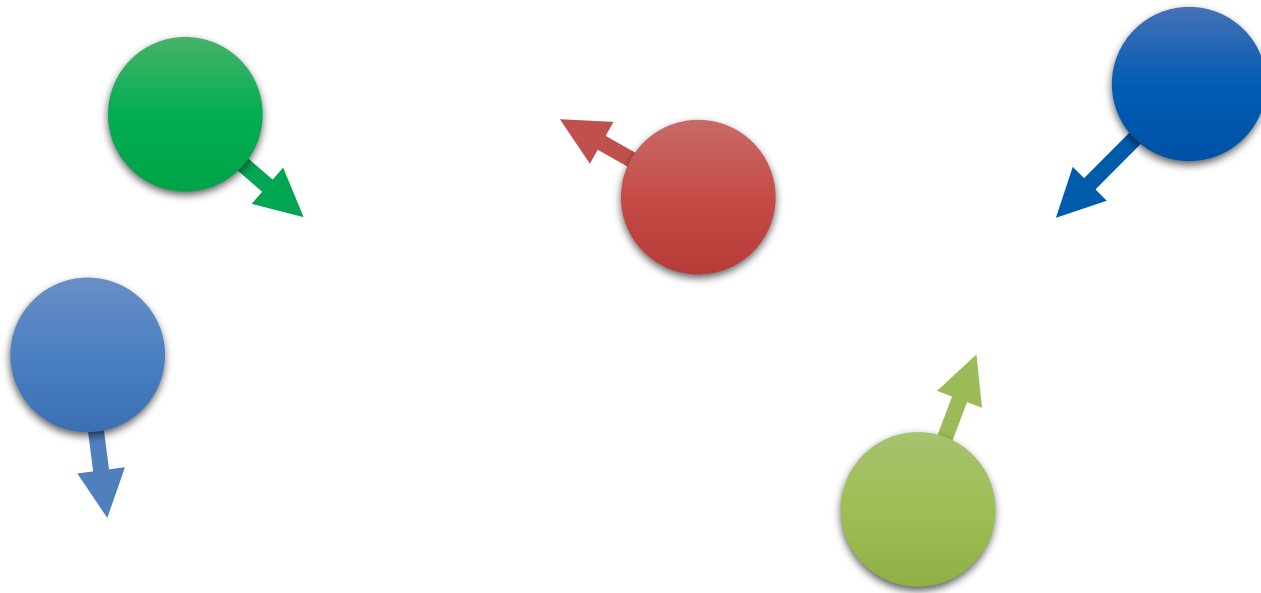
# Nosé-Hoover Thermostat

As the simulation goes on ...



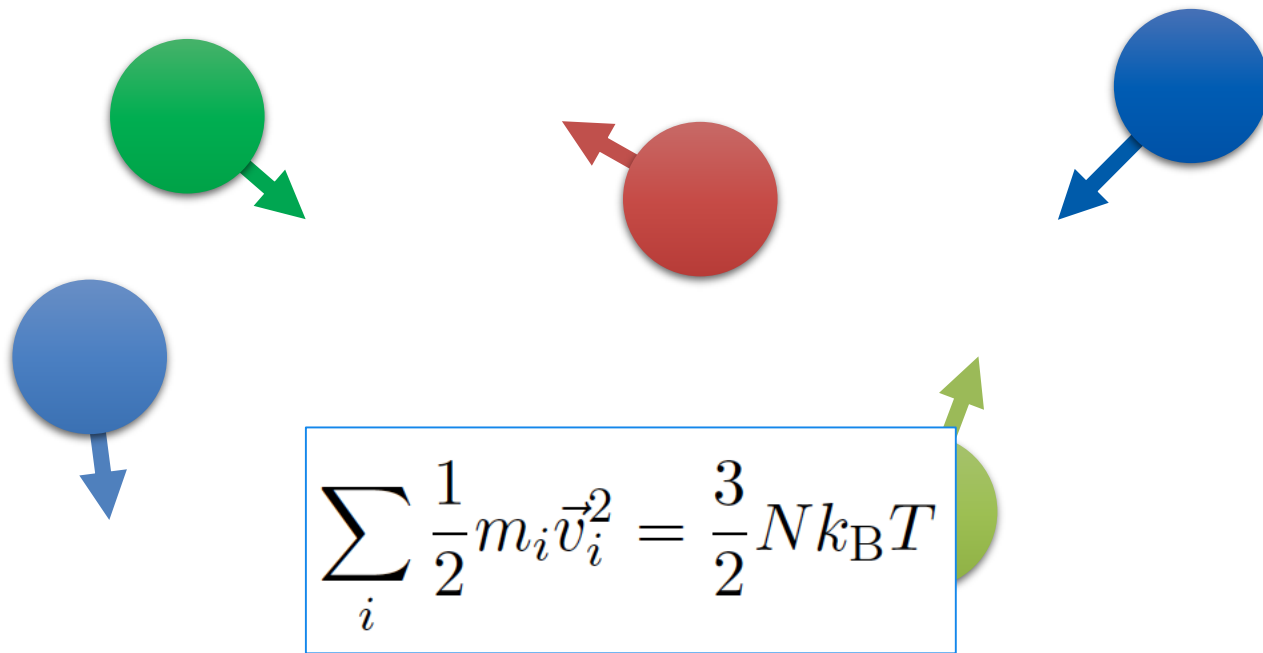
# Nosé-Hoover Thermostat

As the simulation goes on ...

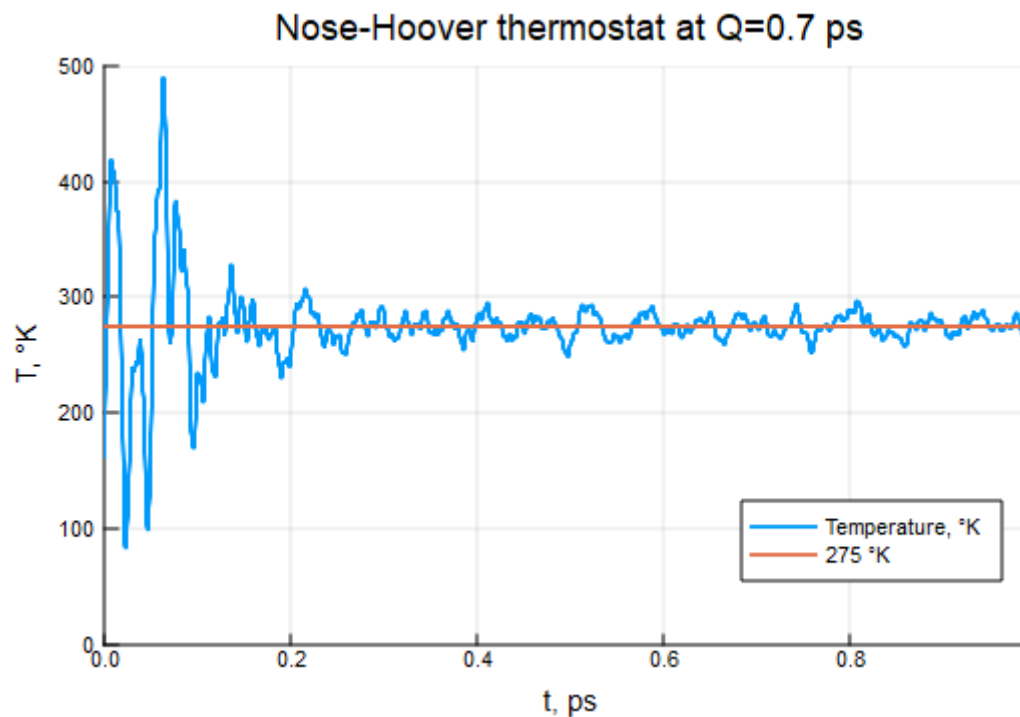


# Nosé-Hoover Thermostat

The target temperature is reached at equilibrium

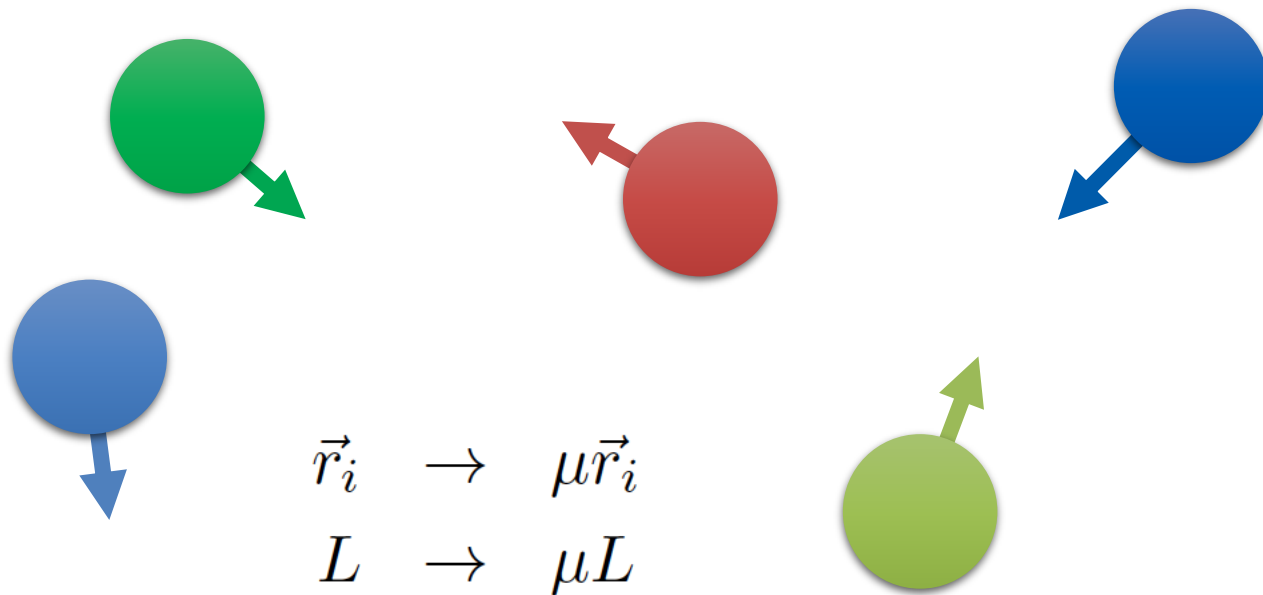


# Nosé-Hoover Thermostat



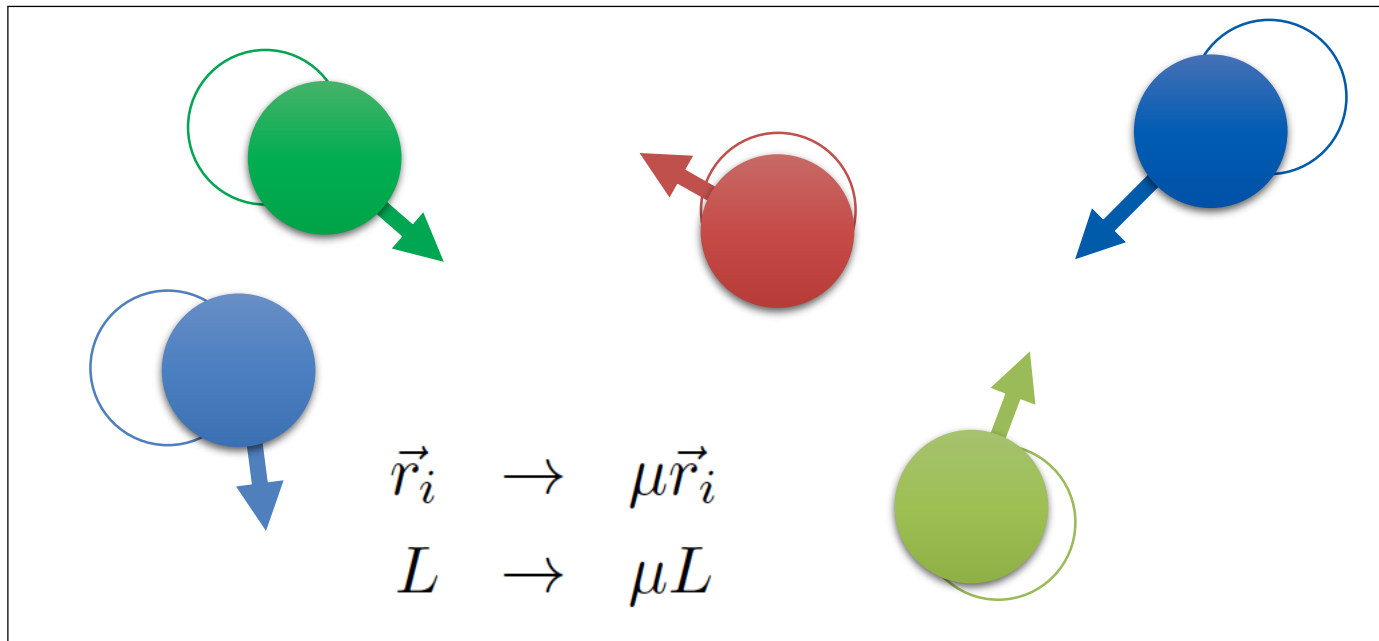
# Berendsen Barostat

Increase/decrease the whole box by a factor of  $\mu$



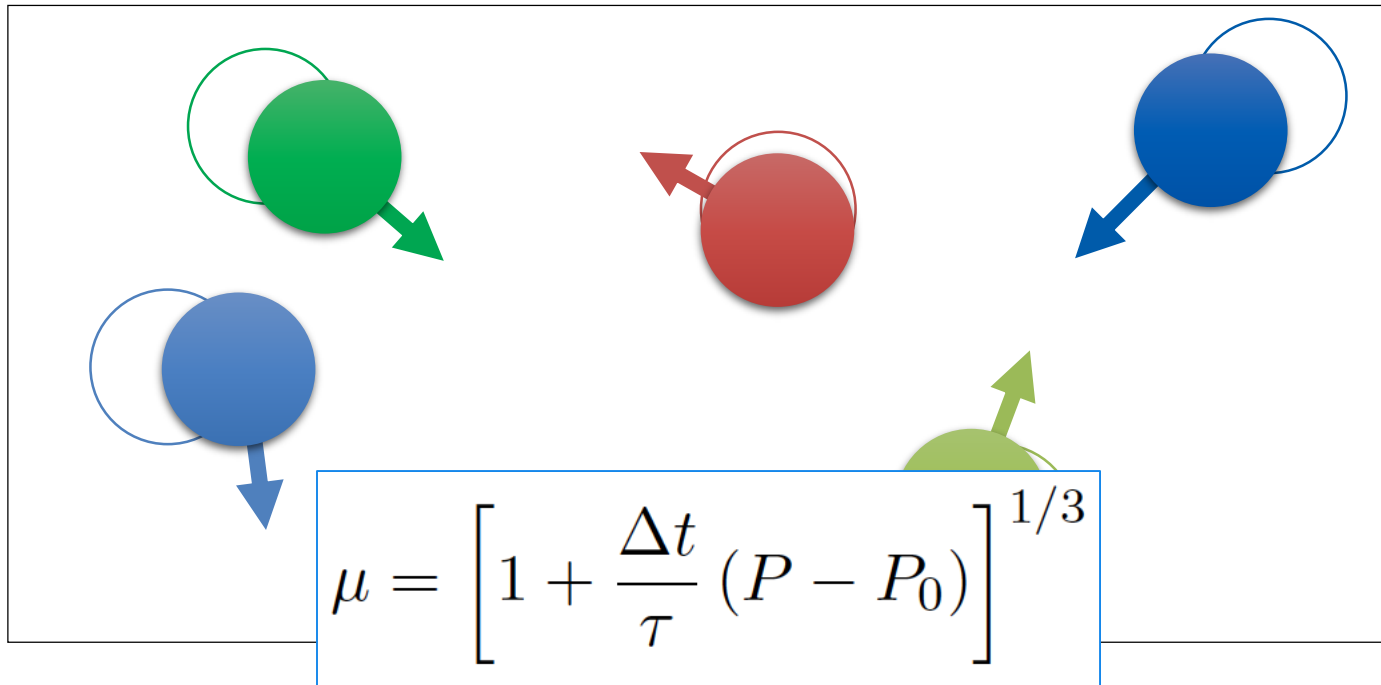
# Berendsen Barostat

Increase/decrease the whole box by a factor of  $\mu$

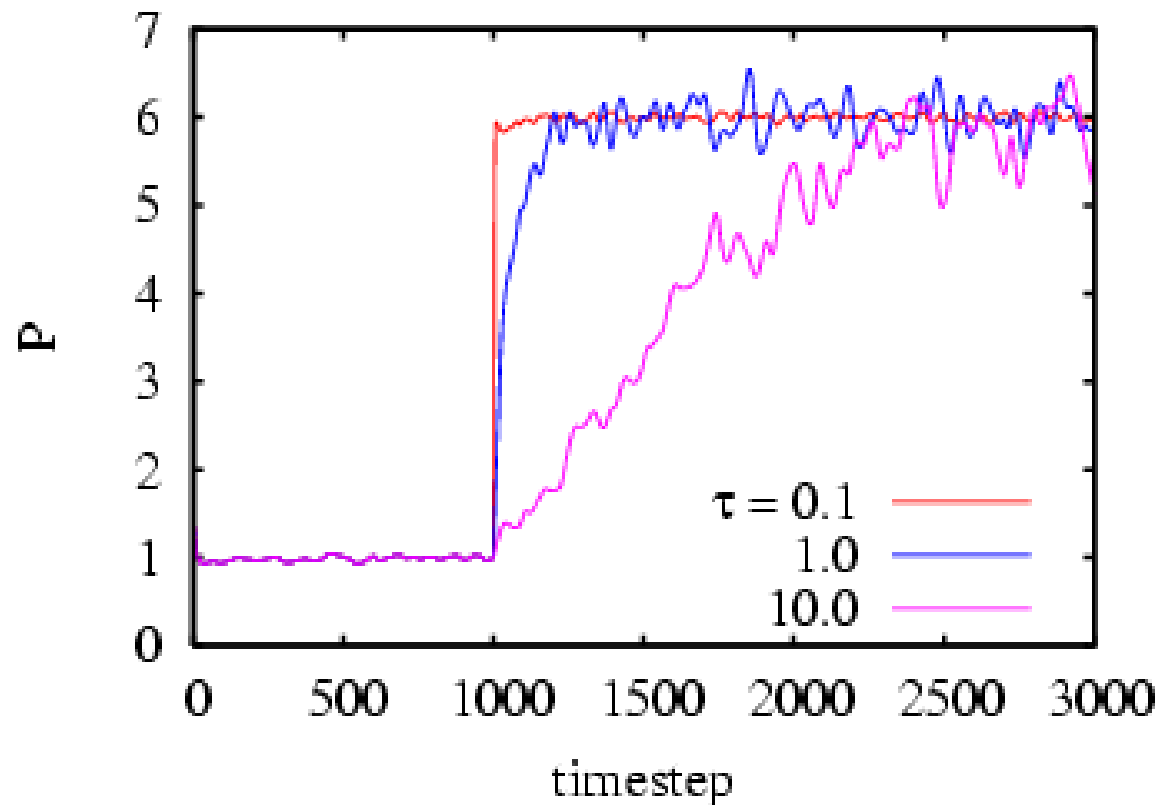


# Berendsen Barostat

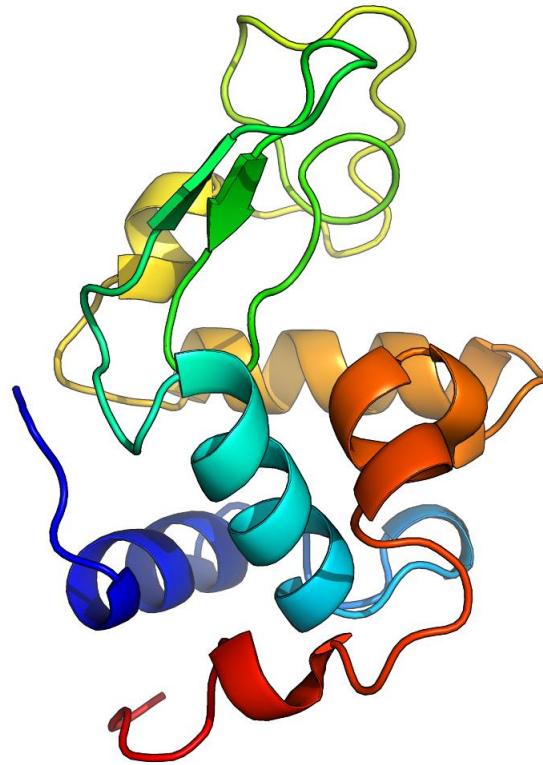
$\mu$  depends on the difference  
between system pressure and target pressure



# Berendsen Barostat



# Tutorial: Lysozyme in Water



# Basic MD Workflow

System preparation

Energy minimization

*NVT* Equilibration

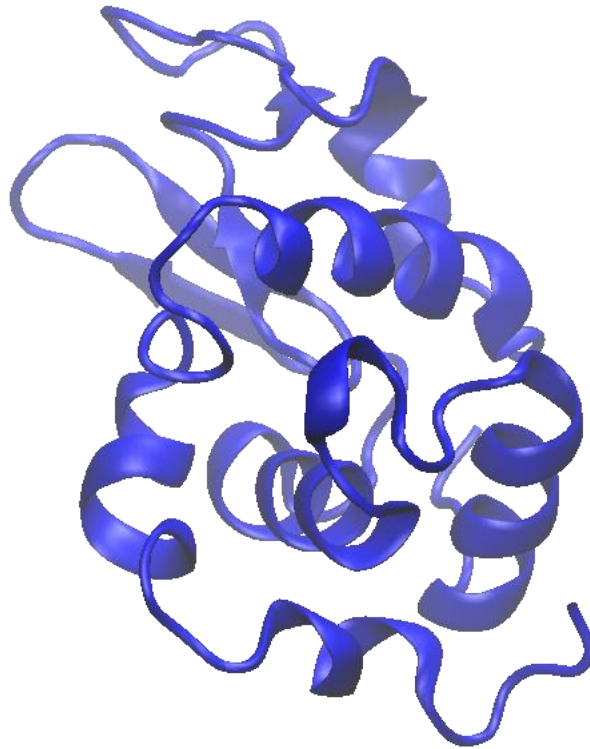
*NPT* Equilibration

Production MD

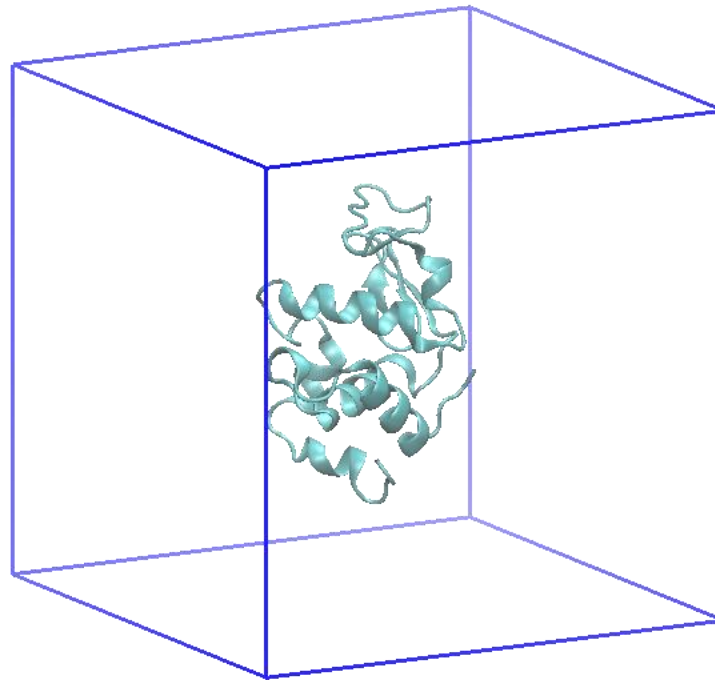
# System Preparation

- What we need:
  - Number and types of particles
  - Interactions: force field
  - Initial positions
  - Initial velocities → randomly set (later)

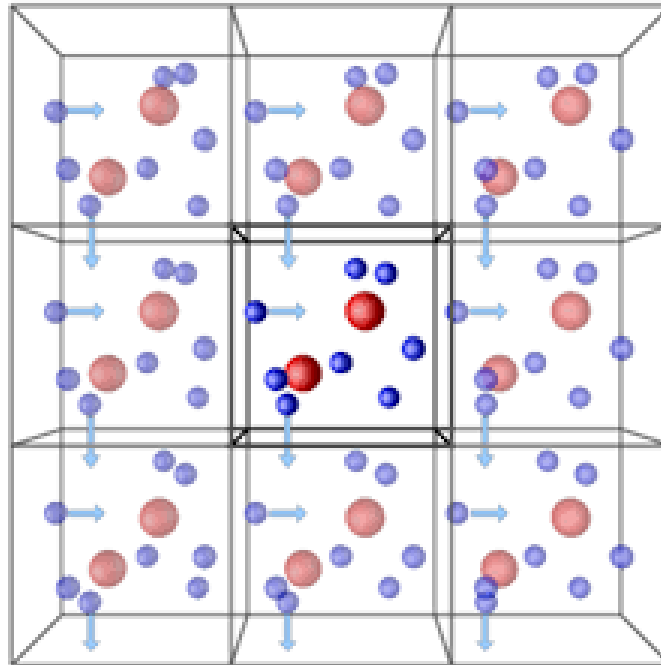
# Experimental Structure



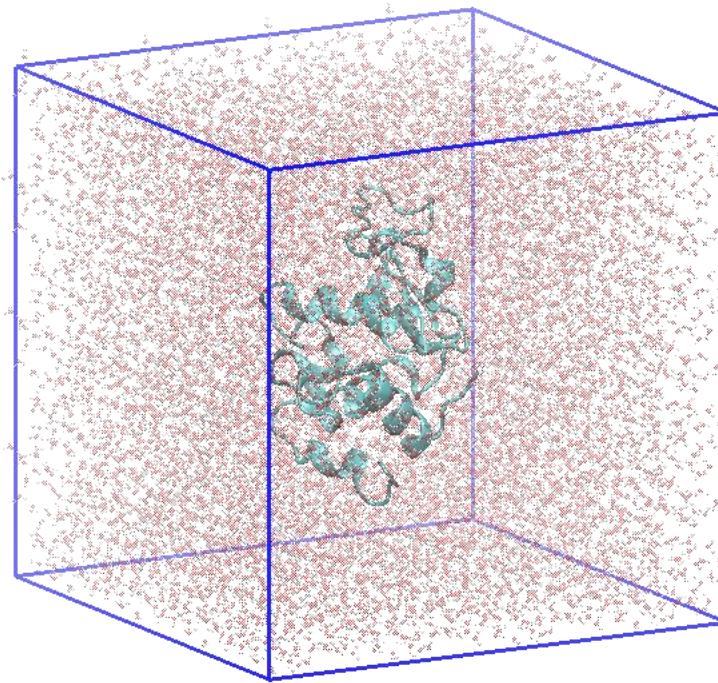
# Simulation Box



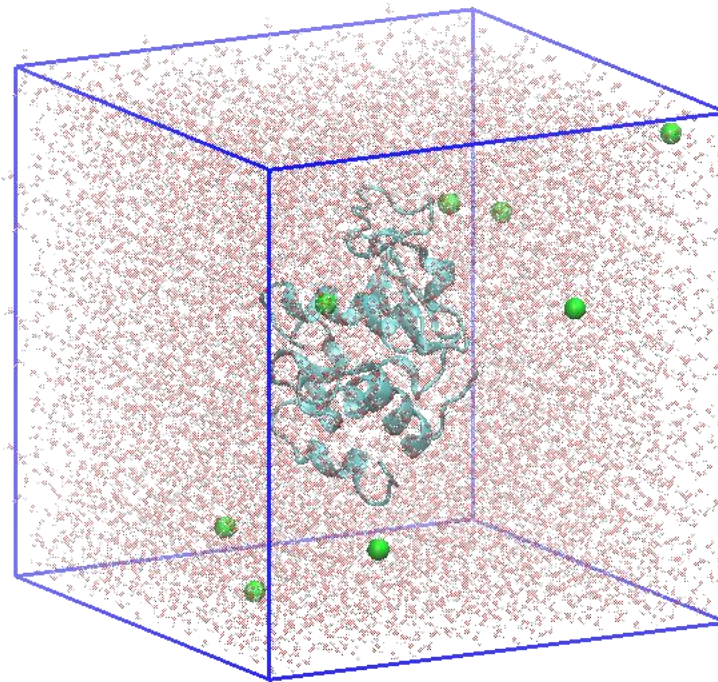
# Periodic Boundary Condition



# Adding Solvent



# Adding Counterions



# Basic MD Workflow

System preparation

Energy minimization

*NVT* Equilibration

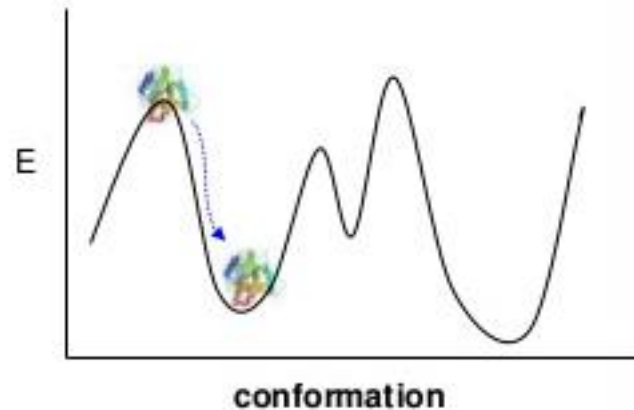
*NPT* Equilibration

Production MD

# Energy Minimization

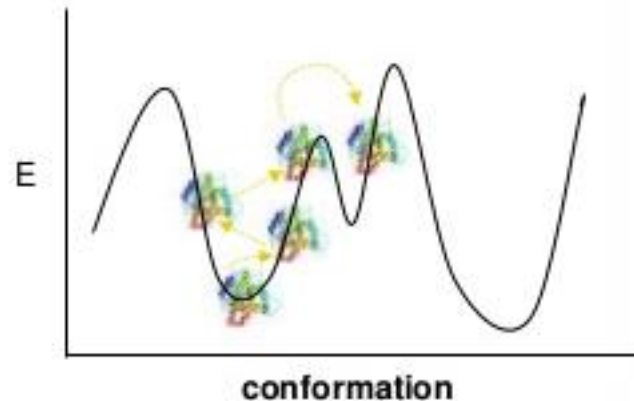
## Minimization

Attempts to find the nearest, low energy conformation



## Molecular dynamics

Seeks to explore conformational space; can access multiple low energy conformations



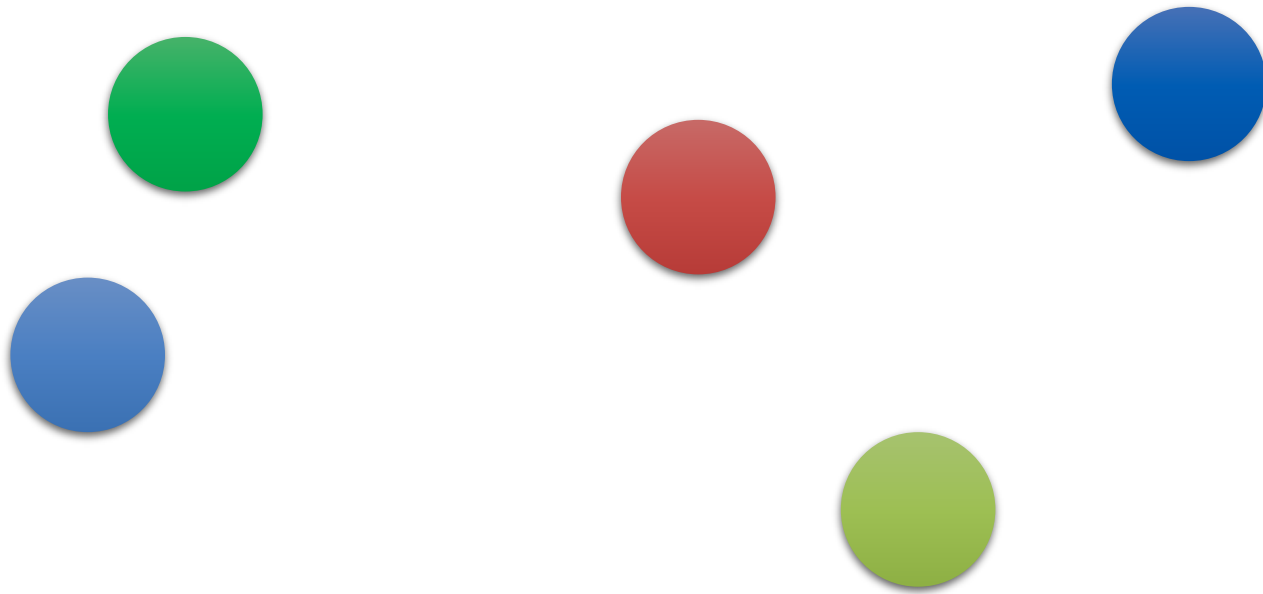
# *NVT* and *NPT* Equilibration

- Gradually add thermostat and barostat
- Avoid abrupt changes
- Start production MD after the system is sufficiently equilibrated

## 2. Monte Carlo Methods

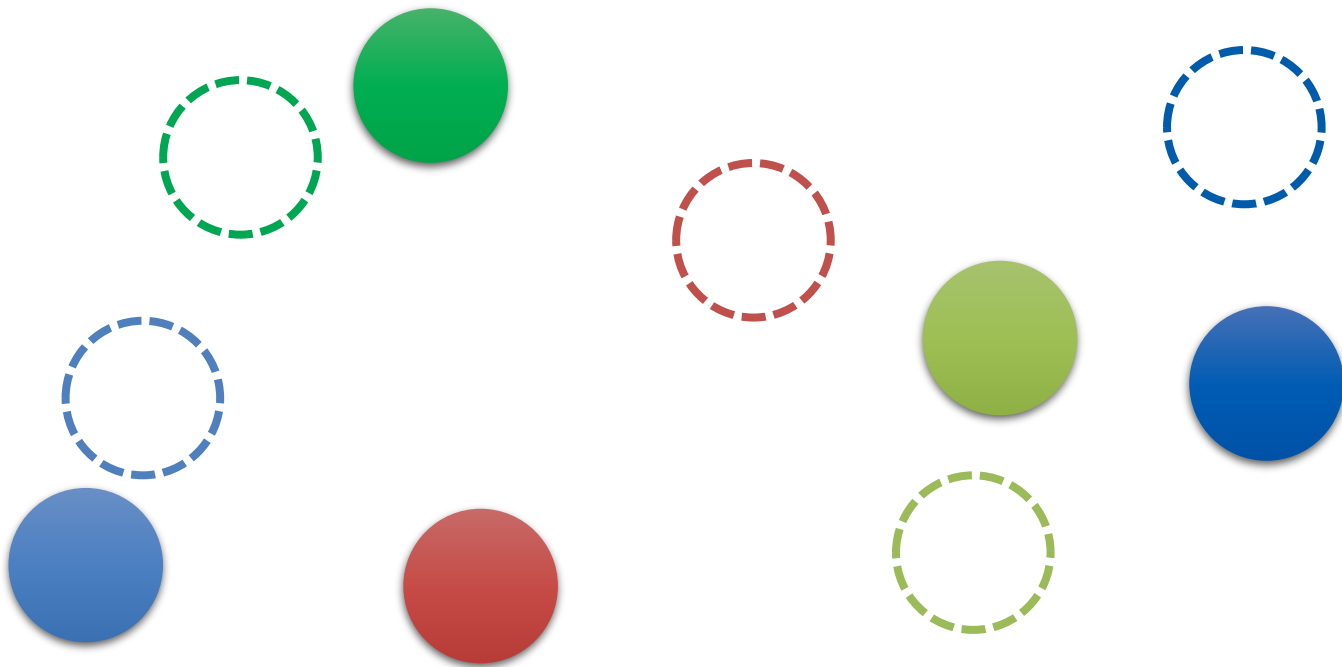


# What is the Monte Carlo Method?



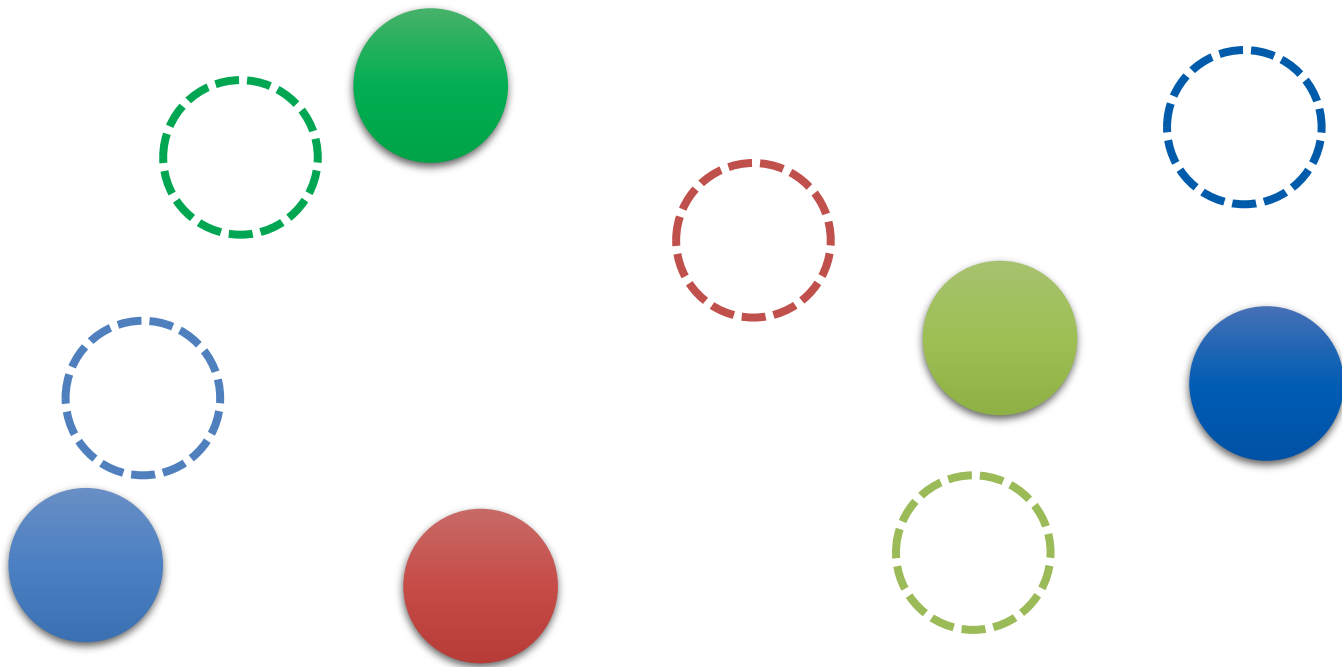
# Monte Carlo (MC) Method

In each MC step, **change configuration**



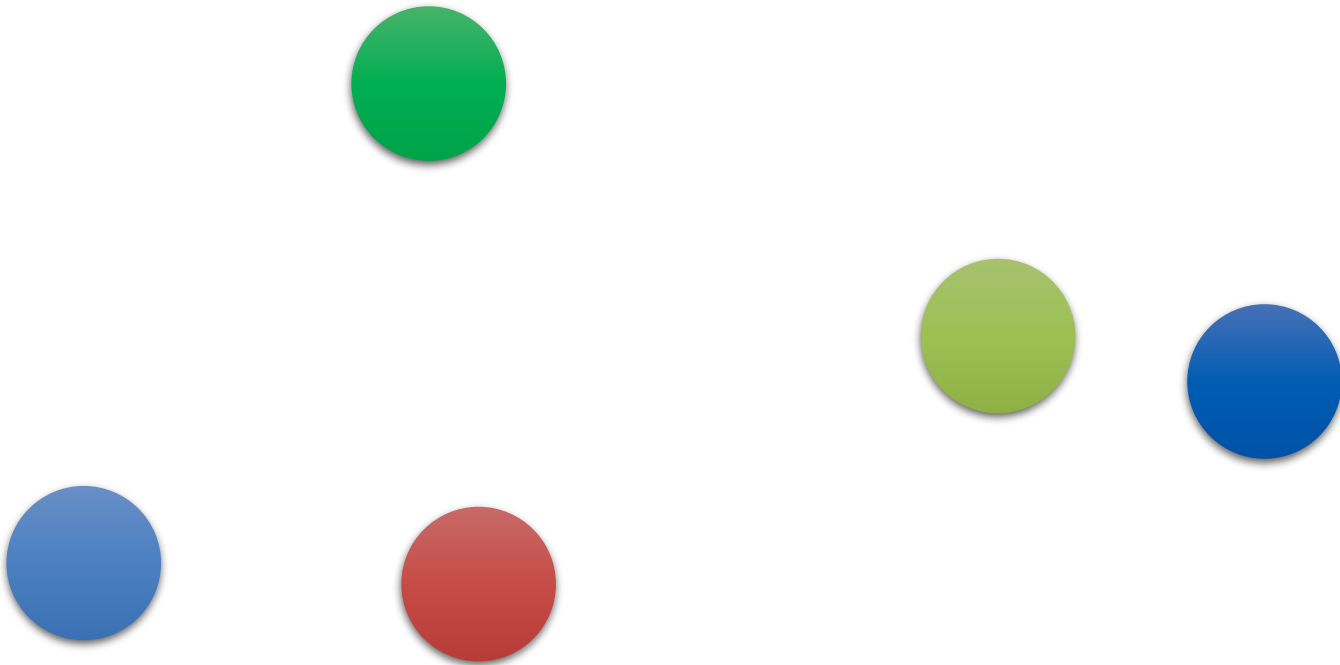
# Monte Carlo (MC) Method

Compare energies of the two config.



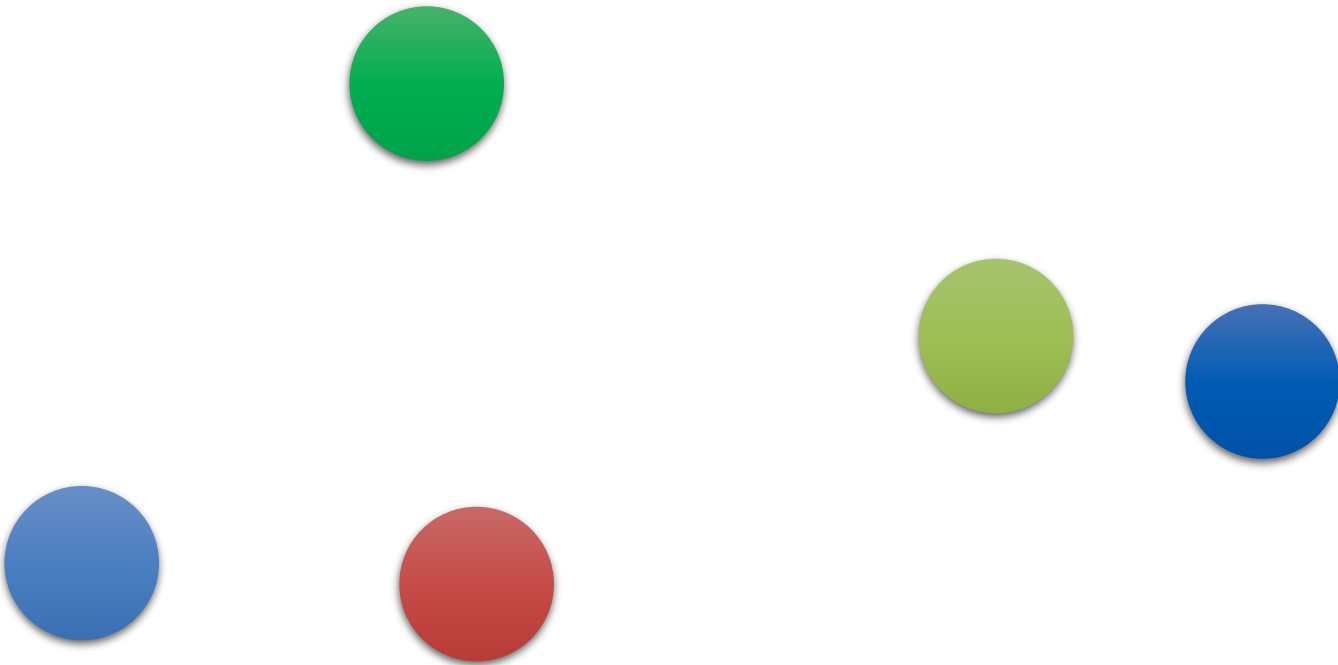
# Monte Carlo (MC) Method

Accept/reject the new configuration based on the energy difference (Metropolis algorithm)



# Monte Carlo (MC) Method

Repeat this again and again



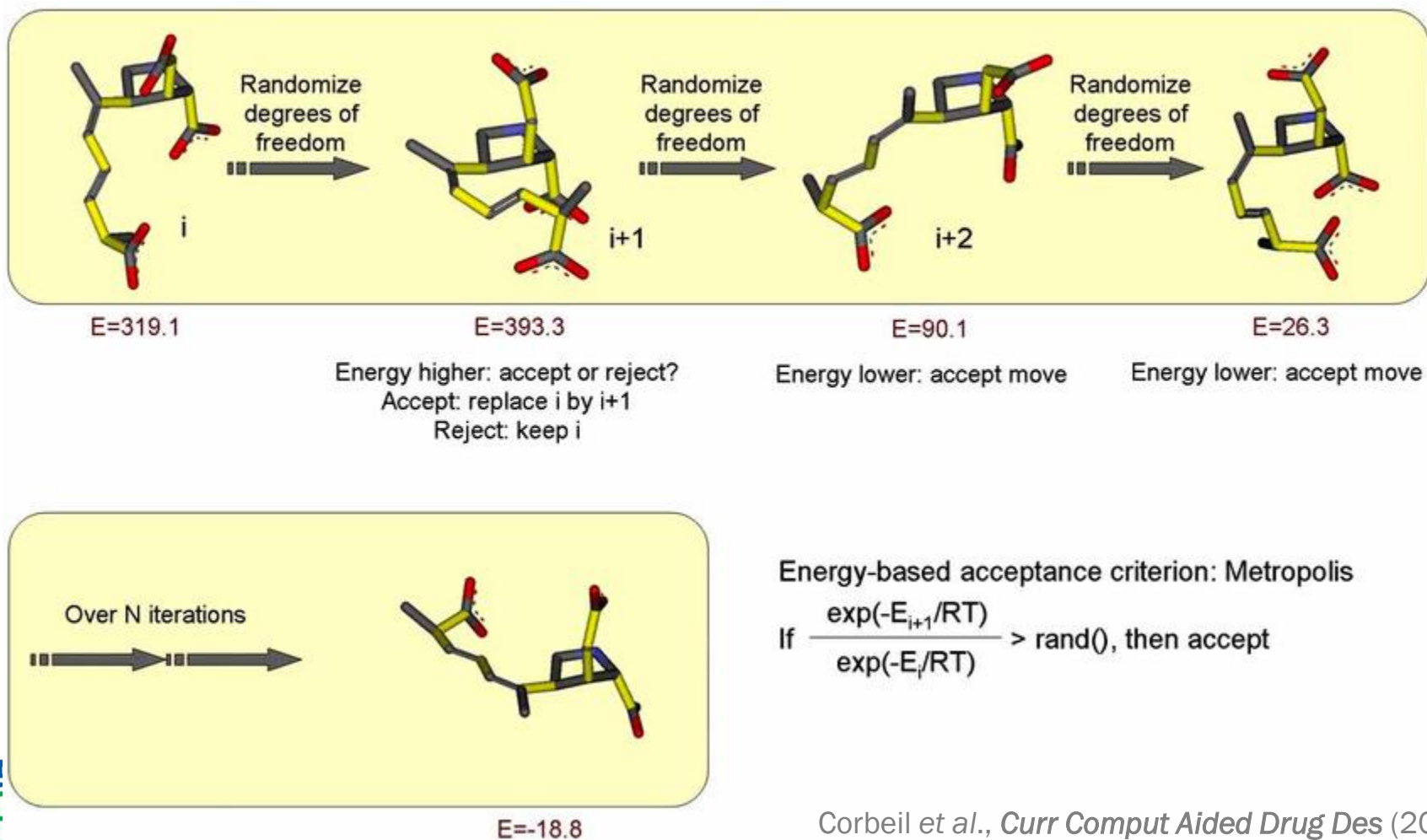
# Metropolis Algorithm

For the energy difference  $\Delta E = E_{\text{trial}} - E_{\text{prev}}$ ,

1. If  $\Delta E \leq 0$ , **accept the move**
2. If  $\Delta E > 0$ , calculate  $P = \exp(-\Delta E/k_B T)$  and draw a random number  $x$  ( $0 \leq x < 1$ )
  - If  $x \leq P$ , **accept the move**
  - If  $x > P$ , **reject the move**



# Example



# Ingredients of MC

- Move set: how to change a configuration
- Potential energy or interactions

# Example: Alloy Phase Separation

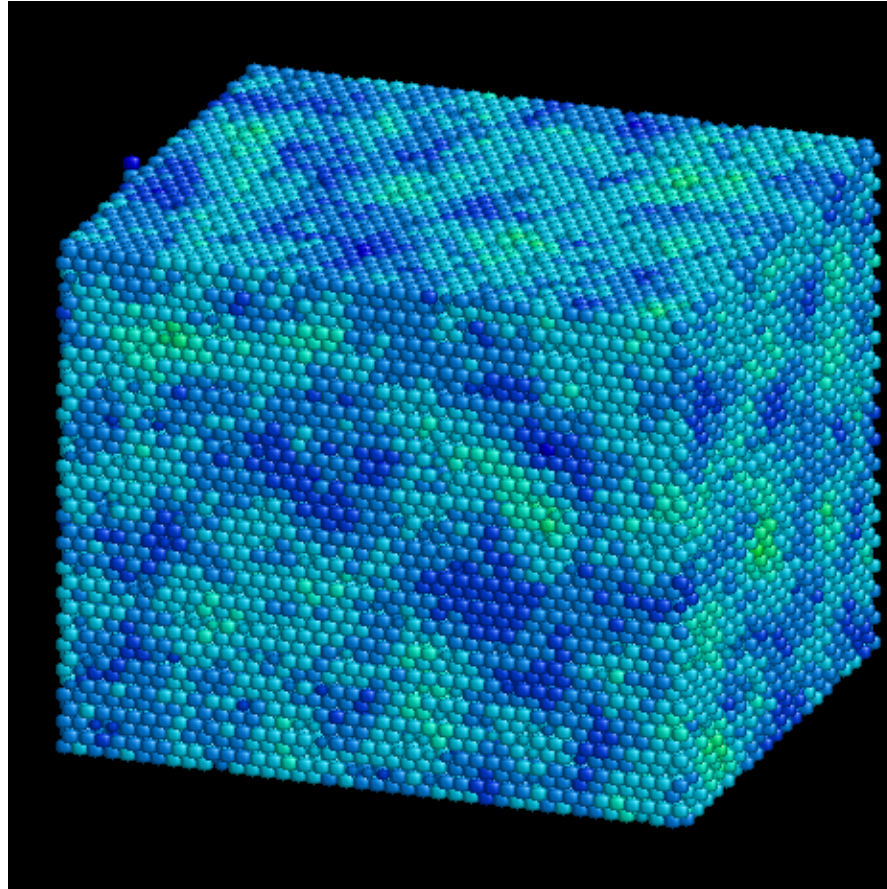
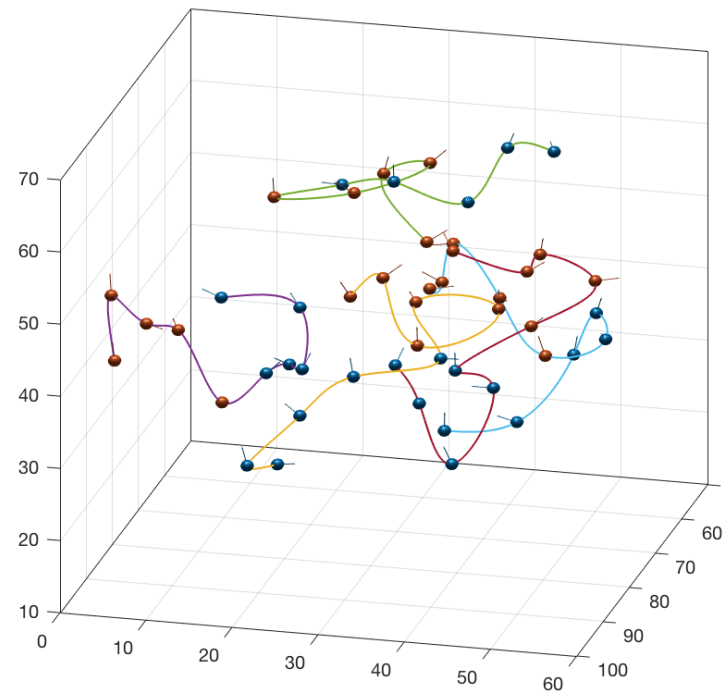
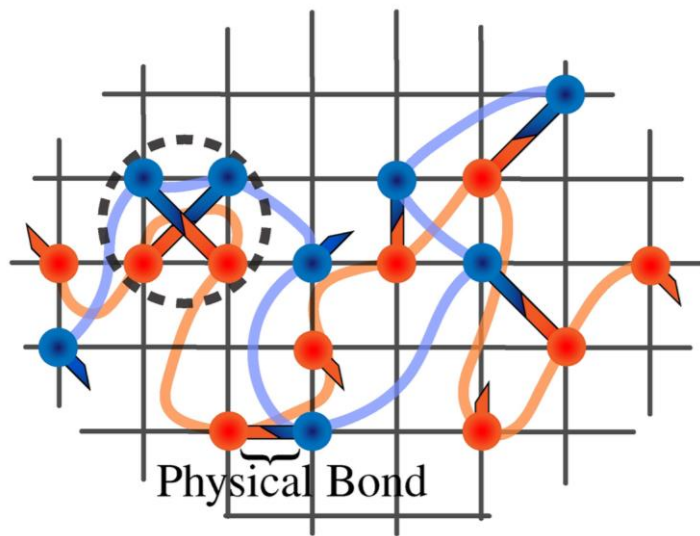
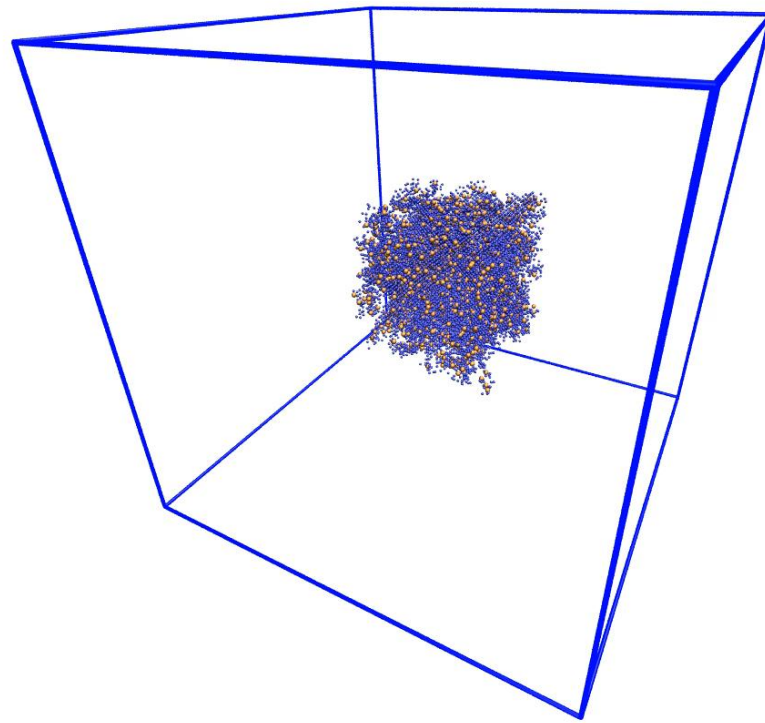


Image: <http://multiscale-cmse.postech.ac.kr/pages/large-scale.html>

# Lattice-Based Simulation



# Example: Polymer Phase Separation



# Comparison: MD vs. MC

## Molecular Dynamics (MD)

In each MD step,

- Calculate force  $F$  on each particle
- Move each particle according to  $F = ma$
- We need timestep  $\Delta t$
- Moves are more *physical*
- Time information

## Monte Carlo (MC)

In each MC step,

- Change configuration
- Compare energies of old and new configurations
- Accept or reject the new configuration based on  $E$
- Moves can be *arbitrary*
- Easy barrier crossing

# Monte Carlo: Not Just Molecules

If you can define **energy function** ...

